

INTRODUCTION TO ELASTICITY

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January 21, 2000

Introduction

This module outlines the basic mechanics of *elastic* response — a physical phenomenon that materials often (but do not always) exhibit. An *elastic* material is one that deforms immediately upon loading, maintains a constant deformation as long as the load is held constant, and returns immediately to its original undeformed shape when the load is removed. This module will also introduce two essential concepts in Mechanics of Materials: *stress* and *strain*.

Tensile strength and tensile stress

Perhaps the most natural test of a material's mechanical properties is the *tension test*, in which a strip or cylinder of the material, having length L and cross-sectional area A , is anchored at one end and subjected to an axial load P — a load acting along the specimen's long axis — at the other. (See Fig. 1). As the load is increased gradually, the axial deflection δ of the loaded end will increase also. Eventually the test specimen breaks or does something else catastrophic, often fracturing suddenly into two or more pieces. (Materials can fail mechanically in many different ways; for instance, recall how blackboard chalk, a piece of fresh wood, and Silly Putty break.) As engineers, we naturally want to understand such matters as how δ is related to P , and what ultimate fracture load we might expect in a specimen of different size than the original one. As materials technologists, we wish to understand how these relationships are influenced by the constitution and microstructure of the material.

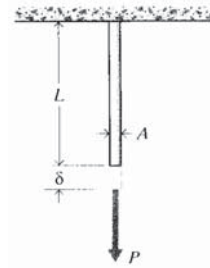


Figure 1: The tension test.

One of the pivotal historical developments in our understanding of material mechanical properties was the realization that the strength of a uniaxially loaded specimen is related to the

magnitude of its *cross-sectional area*. This notion is reasonable when one considers the strength to arise from the number of chemical bonds connecting one cross section with the one adjacent to it as depicted in Fig. 2, where each bond is visualized as a spring with a certain stiffness and strength. Obviously, the number of such bonds will increase proportionally with the section's area¹. The axial strength of a piece of blackboard chalk will therefore increase as the *square* of its diameter. In contrast, increasing the *length* of the chalk will not make it stronger (in fact it will likely become weaker, since the longer specimen will be statistically more likely to contain a strength-reducing flaw.)

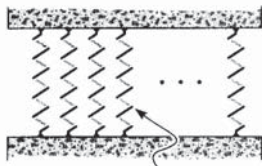


Figure 2: Interplanar bonds (surface density approximately 10^{19} m^{-2}).

Galileo (1564–1642)² is said to have used this observation to note that giants, should they exist, would be very fragile creatures. Their strength would be greater than ours, since the cross-sectional areas of their skeletal and muscular systems would be larger by a factor related to the square of their height (denoted L in the famous DaVinci sketch shown in Fig. 3). But their *weight*, and thus the loads they must sustain, would increase as their volume, that is by the *cube* of their height. A simple fall would probably do them great damage. Conversely, the “proportionate” strength of the famous arachnid mentioned weekly in the *SpiderMan* comic strip is mostly just this same size effect. There’s nothing magical about the muscular strength of insects, but the ratio of L^2 to L^3 works in their favor when strength per body weight is reckoned. This cautions us that simple scaling of a previously proven design is not a safe design procedure. A jumbo jet is not just a small plane scaled up; if this were done the load-bearing components would be too small in cross-sectional area to support the much greater loads they would be called upon to resist.

When reporting the strength of materials loaded in tension, it is customary to account for this effect of area by dividing the breaking load by the cross-sectional area:

$$\boxed{\sigma_f = \frac{P_f}{A_0}} \quad (1)$$

where σ_f is the *ultimate tensile stress*, often abbreviated as UTS, P_f is the load at fracture, and A_0 is the original cross-sectional area. (Some materials exhibit substantial reductions in cross-sectional area as they are stretched, and using the original rather than final area gives the so-call *engineering* strength.) The units of stress are obviously load per unit area, N/m^2 (also

¹The surface density of bonds N_S can be computed from the material’s density ρ , atomic weight W_a and Avogadro’s number N_A as $N_S = (\rho N_A / W_a)^{2/3}$. Illustrating for the case of iron (Fe):

$$N_S = \left(\frac{7.86 \frac{\text{g}}{\text{cm}^3} \cdot 6.023 \times 10^{23} \frac{\text{atoms}}{\text{mol}}}{55.85 \frac{\text{g}}{\text{mol}}} \right)^{\frac{2}{3}} = 1.9 \times 10^{15} \frac{\text{atoms}}{\text{cm}^2}$$

$N_S \approx 10^{15} \frac{\text{atoms}}{\text{cm}^2}$ is true for many materials.

²Galileo, *Two New Sciences*, English translation by H. Crew and A. de Salvio, The Macmillan Co., New York, 1933. Also see S.P. Timoshenko, *History of Strength of Materials*, McGraw-Hill, New York, 1953.

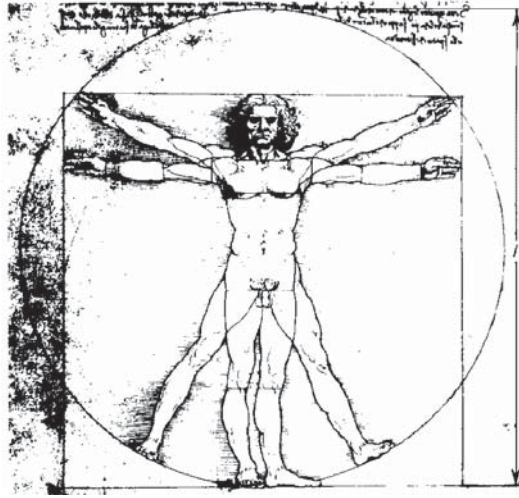


Figure 3: Strength scales with L^2 , but weight scales with L^3 .

called Pascals, or Pa) in the SI system and lb/in² (or psi) in units still used commonly in the United States.

Example 1

In many design problems, the loads to be applied to the structure are known at the outset, and we wish to compute how much material will be needed to support them. As a very simple case, let's say we wish to use a steel rod, circular in cross-sectional shape as shown in Fig. 4, to support a load of 10,000 lb. What should the rod diameter be?

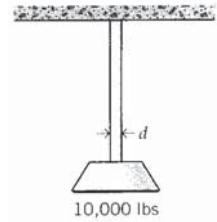


Figure 4: Steel rod supporting a 10,000 lb weight.

Directly from Eqn. 1, the area A_0 that will be just on the verge of fracture at a given load P_f is

$$A_0 = \frac{P_f}{\sigma_f}$$

All we need do is look up the value of σ_f for the material, and substitute it along with the value of 10,000 lb for P_f , and the problem is solved.

A number of materials properties are listed in the *Materials Properties* module, where we find the UTS of carbon steel to be 1200 MPa. We also note that these properties vary widely for given materials depending on their composition and processing, so the 1200 MPa value is only a preliminary design estimate. In light of that uncertainty, and many other potential ones, it is common to include a “factor of safety” in the design. Selection of an appropriate factor is an often-difficult choice, especially in cases where weight or cost restrictions place a great penalty on using excess material. But in this case steel is

relatively inexpensive and we don't have any special weight limitations, so we'll use a conservative 50% safety factor and assume the ultimate tensile strength is $1200/2 = 600$ MPa.

We now have only to adjust the units before solving for area. Engineers must be very comfortable with units conversions, especially given the mix of SI and older traditional units used today. Eventually, we'll likely be ordering steel rod using inches rather than meters, so we'll convert the MPa to psi rather than convert the pounds to Newtons. Also using $A = \pi d^2/4$ to compute the diameter rather than the area, we have

$$d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4P_f}{\pi\sigma_f}} = \left[\frac{4 \times 10000(\text{lb})}{\pi \times 600 \times 10^6(\text{N/m}^2) \times 1.449 \times 10^{-4} \left(\frac{\text{lb/in}^2}{\text{N/m}^2} \right)} \right]^{\frac{1}{2}} = 0.38 \text{ in}$$

We probably wouldn't order rod of exactly 0.38 in, as that would be an oddball size and thus too expensive. But $3/8''$ (0.375 in) would likely be a standard size, and would be acceptable in light of our conservative safety factor.

If the specimen is loaded by an axial force P less than the breaking load P_f , the *tensile stress* is defined by analogy with Eqn. 1 as

$$\sigma = \frac{P}{A_0} \quad (2)$$

The tensile stress, the force per unit area acting on a plane transverse to the applied load, is a fundamental measure of the internal forces within the material. Much of Mechanics of Materials is concerned with elaborating this concept to include higher orders of dimensionality, working out methods of determining the stress for various geometries and loading conditions, and predicting what the material's response to the stress will be.

Example 2

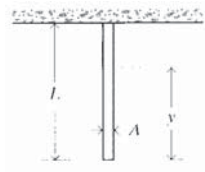


Figure 5: Circular rod suspended from the top and bearing its own weight.

Many engineering applications, notably aerospace vehicles, require materials that are both strong and lightweight. One measure of this combination of properties is provided by computing how long a rod of the material can be that when suspended from its top will break under its own weight (see Fig. 5). Here the stress is not uniform along the rod: the material at the very top bears the weight of the entire rod, but that at the bottom carries no load at all.

To compute the stress as a function of position, let y denote the distance from the bottom of the rod and let the weight density of the material, for instance in N/m^3 , be denoted by γ . (The weight density is related to the mass density ρ [kg/m^3] by $\gamma = \rho g$, where $g = 9.8 \text{ m/s}^2$ is the acceleration due to gravity.) The weight supported by the cross-section at y is just the weight density γ times the volume of material V below y :

$$W(y) = \gamma V = \gamma A y$$

The tensile stress is then given as a function of y by Eqn. 2 as

$$\sigma(y) = \frac{W(y)}{A} = \gamma y$$

Note that the area cancels, leaving only the material density γ as a design variable.

The length of rod that is just on the verge of breaking under its own weight can now be found by letting $y = L$ (the highest stress occurs at the top), setting $\sigma(L) = \sigma_f$, and solving for L :

$$\sigma_f = \gamma L \Rightarrow L = \frac{\sigma_f}{\gamma}$$

In the case of steel, we find the mass density ρ in Appendix A to be $7.85 \times 10^3(\text{kg/m}^3)$; then

$$L = \frac{\sigma_f}{\rho g} = \frac{1200 \times 10^6(\text{N/m}^2)}{7.85 \times 10^3(\text{kg/m}^3) \times 9.8(\text{m/s}^2)} = 15.6 \text{ km}$$

This would be a long rod indeed; the purpose of such a calculation is not so much to design superlong rods as to provide a vivid way of comparing materials (see Prob. 4).

Stiffness

It is important to distinguish *stiffness*, which is a measure of the *load* needed to induce a given *deformation* in the material, from the *strength*, which usually refers to the material's resistance to failure by fracture or excessive deformation. The stiffness is usually measured by applying relatively small loads, well short of fracture, and measuring the resulting deformation. Since the deformations in most materials are very small for these loading conditions, the experimental problem is largely one of measuring small changes in length accurately.

Hooke³ made a number of such measurements on long wires under various loads, and observed that to a good approximation the load P and its resulting deformation δ were related linearly as long as the loads were sufficiently small. This relation, generally known as *Hooke's Law*, can be written algebraically as

$$\boxed{P = k\delta} \tag{3}$$

where k is a constant of proportionality called the *stiffness* and having units of lb/in or N/m. The stiffness as defined by k is not a function of the material alone, but is also influenced by the specimen shape. A wire gives much more deflection for a given load if coiled up like a watch spring, for instance.

A useful way to adjust the stiffness so as to be a purely materials property is to normalize the load by the cross-sectional area; i.e. to use the tensile stress rather than the load. Further, the deformation δ can be normalized by noting that an applied load stretches all parts of the wire uniformly, so that a reasonable measure of "stretching" is the deformation per unit length:

$$\boxed{\epsilon = \frac{\delta}{L_0}} \tag{4}$$

³Robert Hooke (1635–1703) was a contemporary and rival of Isaac Newton. Hooke was a great pioneer in mechanics, but competing with Newton isn't easy.

Here L_0 is the original length and ϵ is a dimensionless measure of stretching called the *strain*. Using these more general measures of load per unit area and displacement per unit length⁴, Hooke's Law becomes:

$$\frac{P}{A_0} = E \frac{\delta}{L_0} \quad (5)$$

or

$$\boxed{\sigma = E\epsilon} \quad (6)$$

The constant of proportionality E , called *Young's modulus*⁵ or the *modulus of elasticity*, is one of the most important mechanical descriptors of a material. It has the same units as stress, Pa or psi. As shown in Fig. 6, Hooke's law can refer to either of Eqns. 3 or 6.

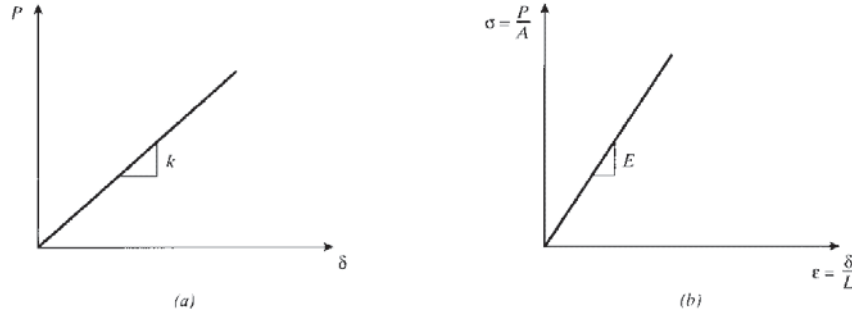


Figure 6: Hooke's law in terms of (a) load-displacement and (b) stress-strain.

The Hookean stiffness k is now recognizable as being related to the Young's modulus E and the specimen geometry as

$$k = \frac{AE}{L} \quad (7)$$

where here the 0 subscript is dropped from the area A ; it will be assumed from here on (unless stated otherwise) that the change in area during loading can be neglected. Another useful relation is obtained by solving Eqn. 5 for the deflection in terms of the applied load as

$$\boxed{\delta = \frac{PL}{AE}} \quad (8)$$

Note that the stress $\sigma = P/A$ developed in a tensile specimen subjected to a fixed load is independent of the material properties, while the deflection depends on the material property E . Hence the stress σ in a tensile specimen at a given load is the same whether it's made of steel or polyethylene, but the strain ϵ would be different: the polyethylene will exhibit much larger strain and deformation, since its modulus is two orders of magnitude less than steel's.

⁴It was apparently the Swiss mathematician Jakob Bernoulli (1655-1705) who first realized the correctness of this form, published in the final paper of his life.

⁵After the English physicist Thomas Young (1773–1829), who also made notable contributions to the understanding of the interference of light as well as being a noted physician and Egyptologist.

Example 3

In Example 1, we found that a steel rod 0.38" in diameter would safely bear a load of 10,000 lb. Now let's assume we have been given a second design goal, namely that the geometry requires that we use a rod 15 ft in length but that the loaded end cannot be allowed to deflect downward more than 0.3" when the load is applied. Replacing A in Eqn. 8 by $\pi d^2/4$ and solving for d , the diameter for a given δ is

$$d = 2\sqrt{\frac{PL}{\pi\delta E}}$$

From Appendix A, the modulus of carbon steel is 210 GPa; using this along with the given load, length, and deflection, the required diameter is

$$d = 2\sqrt{\frac{10^4(\text{lb}) \times 15(\text{ft}) \times 12(\text{in}/\text{ft})}{\pi \times 0.3(\text{in}) \times 210 \times 10^9(\text{N}/\text{m}^2) \times 1.449 \times 10^{-4} \left(\frac{\text{lb}/\text{in}^2}{\text{N}/\text{m}^2}\right)}} = 0.5 \text{ in}$$

This diameter is larger than the 0.38" computed earlier; therefore a larger rod must be used if the deflection as well as the strength goals are to be met. Clearly, using the larger rod makes the tensile stress in the material less and thus lowers the likelihood of fracture. This is an example of a *stiffness-critical* design, in which deflection rather than fracture is the governing constraint. As it happens, many structures throughout the modern era have been designed for stiffness rather than strength, and thus wound up being "overdesigned" with respect to fracture. This has undoubtedly lessened the incidence of fracture-related catastrophes, which will be addressed in the modules on fracture.

Example 4

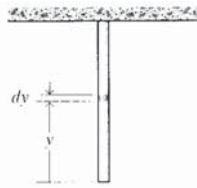


Figure 7: Deformation of a column under its own weight.

When very long columns are suspended from the top, as in a cable hanging down the hole of an oil well, the deflection due to the weight of the material itself can be important. The solution for the total deflection is a minor extension of Eqn. 8, in that now we must consider the increasing weight borne by each cross section as the distance from the bottom of the cable increases. As shown in Fig. 7, the total elongation of a column of length L , cross-sectional area A , and weight density γ due to its own weight can be found by considering the incremental deformation $d\delta$ of a slice dy a distance y from the bottom. The weight borne by this slice is γAy , so

$$d\delta = \frac{(\gamma Ay) dy}{AE}$$

$$\delta = \int_0^L d\delta = \frac{\gamma}{E} \frac{y^2}{2} \Big|_0^L = \frac{\gamma L^2}{2E}$$

Note that δ is independent of the area A , so that finding a fatter cable won't help to reduce the deformation; the critical parameter is the *specific modulus* E/γ . Since the total weight is $W = \gamma AL$, the result can also be written

$$\delta = \frac{WL}{2AE}$$

The deformation is the same as in a bar being pulled with a tensile force equal to half its weight; this is just the average force experienced by cross sections along the column.

In Example 2, we computed the length of a steel rod that would be just on the verge of breaking under its own weight if suspended from its top; we obtained $L = 15.6\text{km}$. Were such a rod to be constructed, our analysis predicts the deformation at the bottom would be

$$\delta = \frac{\gamma L^2}{2E} = \frac{7.85 \times 10^3(\text{kg/m}^3) \times 9.8(\text{m/s}^2) \times [15.6 \times 10^3(\text{m})]^2}{2 \times 210 \times 10^9(\text{N/m}^2)} = 44.6 \text{ m}$$

However, this analysis assumes Hooke's law holds over the entire range of stresses from zero to fracture. This is not true for many materials, including carbon steel, and later modules will address materials response at high stresses.

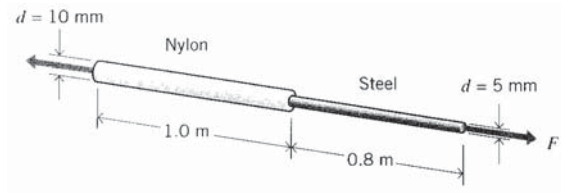
A material that obeys Hooke's Law (Eqn. 6) is called *Hookean*. Such a material is *elastic* according to the description of elasticity given in the introduction (immediate response, full recovery), and it is also *linear* in its relation between stress and strain (or equivalently, force and deformation). Therefore a Hookean material is *linear elastic*, and materials engineers use these descriptors interchangeably. It is important to keep in mind that not all elastic materials are linear (rubber is elastic but nonlinear), and not all linear materials are elastic (viscoelastic materials can be linear in the mathematical sense, but do not respond immediately and are thus not elastic).

The linear proportionality between stress and strain given by Hooke's law is not nearly as general as, say, Einstein's general theory of relativity, or even Newton's law of gravitation. It's really just an approximation that is observed to be reasonably valid for many materials as long the applied stresses are not too large. As the stresses are increased, eventually more complicated material response will be observed. Some of these effects will be outlined in the *Module on Stress-Strain Curves*, which introduces the experimental measurement of the strain response of materials over a range of stresses up to and including fracture.

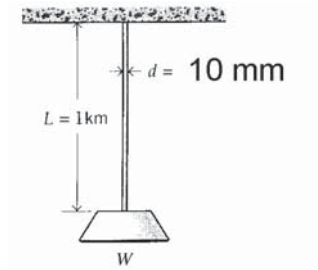
If we were to push on the specimen rather than pulling on it, the loading would be described as *compressive* rather than tensile. In the range of relatively low loads, Hooke's law holds for this case as well. By convention, compressive stresses and strains are negative, so the expression $\sigma = E\epsilon$ holds for both tension and compression.

Problems

1. Determine the stress and total deformation of an aluminum wire, 30 m long and 5 mm in diameter, subjected to an axial load of 250 N.
2. Two rods, one of nylon and one of steel, are rigidly connected as shown. Determine the stresses and axial deformations when an axial load of $F = 1 \text{ kN}$ is applied.
3. A steel cable 10 mm in diameter and 1 km long bears a load in addition to its own weight of $W = 150 \text{ N}$. Find the total elongation of the cable.
4. Using the numerical values given in the *Module on Material Properties*, rank the given materials in terms of the length of rod that will just barely support its own weight.
5. Plot the maximum self-supporting rod lengths of the materials in Prob. 4 versus the cost (per unit cross-sectional area) of the rod.



Prob. 2

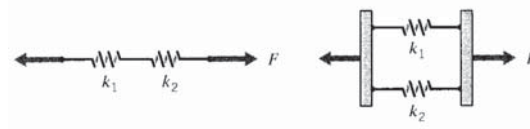


Prob. 3

6. Show that the effective stiffnesses of two springs connected in (a) series and (b) parallel is

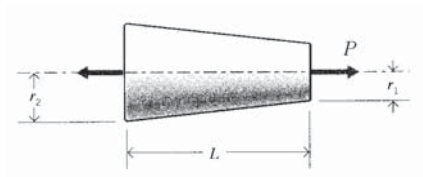
$$(a) \text{ series : } \frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} \quad (b) \text{ parallel : } k_{eff} = k_1 + k_2$$

(Note that these are the reverse of the relations for the effective electrical resistance of two resistors connected in series and parallel, which use the same symbols.)

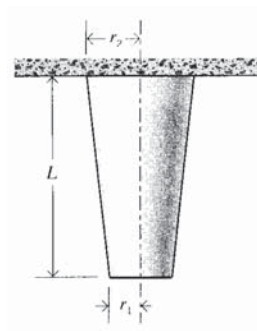


Prob. 6

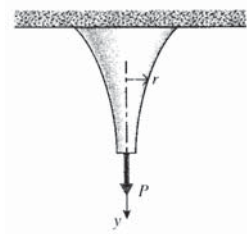
7. A tapered column of modulus E and mass density ρ varies linearly from a radius of r_1 to r_2 in a length L . Find the total deformation caused by an axial load P .
8. A tapered column of modulus E and mass density ρ varies linearly from a radius of r_1 to r_2 in a length L , and is hanging from its broad end. Find the total deformation due to the weight of the bar.
9. A rod of circular cross section hangs under the influence of its own weight, and also has an axial load P suspended from its free end. Determine the shape of the bar, i.e. the function $r(y)$ such that the axial stress is constant along the bar's length.
10. A bolt with 20 threads per inch passes through a sleeve, and a nut is threaded over the bolt as shown. The nut is then tightened one half turn beyond finger tightness; find the stresses in the bolt and the sleeve. All materials are steel, the cross-sectional area of the bolt is 0.5 in^2 , and the area of the sleeve is 0.4 in^2 .



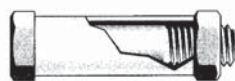
Prob. 7



Prob. 8



Prob. 9



Prob. 10

ATOMISTIC BASIS OF ELASTICITY

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January 27, 2000

Introduction

The *Introduction to Elastic Response* Module introduced two very important material properties, the ultimate tensile strength σ_f and the Young's modulus E . To the effective mechanical designer, these aren't just numerical parameters that are looked up in tables and plugged into equations. The very nature of the material is reflected in these properties, and designers who try to function without a sense of how the material really works are very apt to run into trouble. Whenever practical in these modules, we'll make an effort to put the material's mechanical properties in context with its processing and microstructure. This module will describe how for most engineering materials the modulus is controlled by the atomic bond energy function.

For most materials, the amount of stretching experienced by a tensile specimen under a small fixed load is controlled in a relatively simple way by the tightness of the chemical bonds at the atomic level, and this makes it possible to relate stiffness to the chemical architecture of the material. This is in contrast to more complicated mechanical properties such as fracture, which are controlled by a diverse combination of microscopic as well as molecular aspects of the material's internal structure and surface. Further, the stiffness of some materials — notably rubber — arises not from bond stiffness but from disordering or entropic factors. Some principal aspects of these atomistic views of elastic response are outlined in the sections to follow.

Energetic effects

Chemical bonding between atoms can be viewed as arising from the electrostatic attraction between regions of positive and negative electronic charge. Materials can be classified based on the nature of these electrostatic forces, the three principal classes being

1. *Ionic materials*, such as NaCl, in which an electron is transferred from the less electronegative element (Na) to the more electronegative (Cl). The ions therefore differ by one electronic charge and are thus attracted to one another. Further, the two ions feel an attraction not only to each other but also to other oppositely charged ions in their vicinity; they also feel a repulsion from nearby ions of the same charge. Some ions may gain or lose more than one electron.
2. *Metallic materials*, such as iron and copper, in which one or more loosely bound outer electrons are released into a common pool which then acts to bind the positively charged atomic cores.

3. Covalent materials, such as diamond and polyethylene, in which atomic orbitals overlap to form a region of increased electronic charge to which both nuclei are attracted. This bond is directional, with each of the nuclear partners in the bond feeling an attraction to the negative region between them but not to any of the other atoms nearby.

In the case of ionic bonding, Coulomb's law of electrostatic attraction can be used to develop simple but effective relations for the bond stiffness. For ions of equal charge e the attractive force f_{attr} can be written:

$$f_{attr} = \frac{Ce^2}{r^2} \quad (1)$$

Here C is a conversion factor; For e in Coulombs, $C = 8.988 \times 10^9 \text{ N-m}^2/\text{Coul}^2$. For singly ionized atoms, $e = 1.602 \times 10^{-19} \text{ Coul}$ is the charge on an electron. The *energy* associated with the Coulombic attraction is obtained by integrating the force, which shows that the bond energy varies inversely with the separation distance:

$$U_{attr} = \int f_{attr} dr = \frac{-Ce^2}{r} \quad (2)$$

where the energy of atoms at infinite separation is taken as zero.

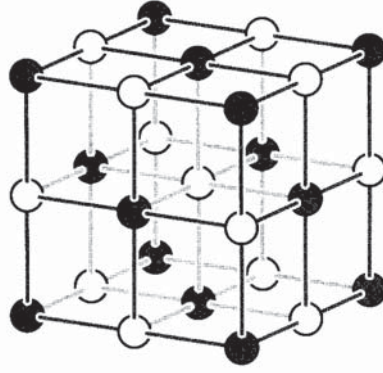


Figure 1: The interpenetrating cubic NaCl lattice.

If the material's atoms are arranged as a perfect crystal, it is possible to compute the electrostatic binding energy field in considerable detail. In the interpenetrating cubic lattice of the ionic NaCl structure shown in Fig. 1, for instance, each ion feels attraction to oppositely charged neighbors and repulsion from equally charged ones. A particular sodium atom is surrounded by 6 Cl^- ions at a distance r , 12 Na^+ ions at a distance $r\sqrt{2}$, 8 Cl^- ions at a distance $r\sqrt{3}$, etc. The total electronic field sensed by the first sodium ion is then:

$$\begin{aligned} U_{attr} &= -\frac{Ce^2}{r} \left(\frac{6}{\sqrt{1}} - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \frac{6}{\sqrt{4}} + \frac{24}{\sqrt{5}} - \dots \right) \\ &= \frac{-ACe^2}{r} \end{aligned} \quad (3)$$

where $A = 1.747558 \dots$ is the result of the previous series, called the *Madelung constant*¹. Note that it is not sufficient to consider only nearest-neighbor attractions in computing the bonding energy; in fact the second term in the series is larger in magnitude than the first. The specific value for the Madelung constant is determined by the crystal structure, being 1.763 for CsCl and 1.638 for cubic ZnS.

At close separation distances, the attractive electrostatic force is balanced by mutual repulsion forces that arise from interactions between overlapping electron shells of neighboring ions; this force varies much more strongly with the distance, and can be written:

$$U_{rep} = \frac{B}{r^n} \quad (4)$$

Compressibility experiments have determined the exponent n to be 7.8 for the NaCl lattice, so this is a much steeper function than U_{attr} .

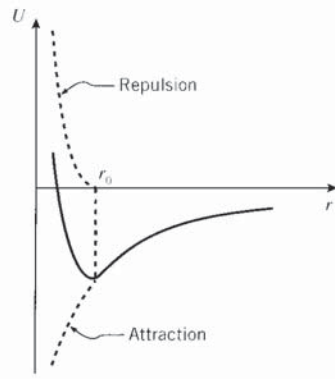


Figure 2: The bond energy function.

As shown in Fig. 2, the total binding energy of one ion due to the presence of all others is then the sum of the attractive and repulsive components:

$$U = -\frac{ACe^2}{r} + \frac{B}{r^n} \quad (5)$$

Note that the curve is *anharmonic* (not shaped like a sine curve), being more flattened out at larger separation distances. The system will adopt a configuration near the position of lowest energy, computed by locating the position of zero slope in the energy function:

$$\begin{aligned} (f)_{r=r_0} &= \left(\frac{dU}{dr} \right)_{r=r_0} = \left(\frac{ACe^2}{r^2} - \frac{nB}{r^{n+1}} \right)_{r=r_0} = 0 \\ r_0 &= \left(\frac{nB}{ACe^2} \right)^{\frac{1}{n-1}} \end{aligned} \quad (6)$$

The range for n is generally 5–12, increasing as the number of outer-shell electrons that cause the repulsive force.

¹C. Kittel, *Introduction to Solid State Physics*, John Wiley & Sons, New York, 1966. The Madelung series does not converge smoothly, and this text includes some approaches to computing the sum.

Example 1

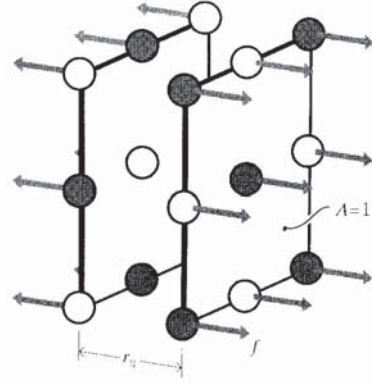


Figure 3: Simple tension applied to crystal face.

In practice the n and B parameters in Eqn. 5 are determined from experimental measurements, for instance by using a combination of X-ray diffraction to measure r_0 and elastic modulus to infer the slope of the $U(r)$ curve. As an illustration of this process, picture a tensile stress σ applied to a unit area of crystal ($A = 1$) as shown in Fig. 3, in a direction perpendicular to the crystal cell face. (The $[100]$ direction on the (100) face, using crystallographic notation².) The total force on this unit area is numerically equal to the stress: $F = \sigma A = \sigma$.

If the interionic separation is r_0 , there will be $1/r_0^2$ ions on the unit area, each being pulled by a force f . Since the total force F is just f times the number of ions, the stress can then be written

$$\sigma = F = f \frac{1}{r_0^2}$$

When the separation between two adjacent ions is increased by an amount δ , the strain is $\epsilon = \delta/r_0$. The differential strain corresponding to a differential displacement is then

$$d\epsilon = \frac{dr}{r_0}$$

The elastic modulus E is now the ratio of stress to strain, in the limit as the strain approaches zero:

$$E = \left. \frac{d\sigma}{d\epsilon} \right|_{\epsilon \rightarrow 0} = \left. \frac{1}{r_0} \frac{df}{dr} \right|_{r \rightarrow r_0} = \frac{1}{r_0} \frac{d}{dr} \left(\frac{ACe^2}{r^2} - \frac{nB}{r^{n+1}} \right) \bigg|_{r \rightarrow r_0}$$

Using $B = ACe^2 r_0^{n-1}/n$ from Eqn. 6 and simplifying,

$$E = \frac{(n-1)ACe^2}{r_0^4}$$

Note the very strong dependence of E on r_0 , which in turn reflects the tightness of the bond. If E and r_0 are known experimentally, n can be determined. For NaCl, $E = 3 \times 10^{10}$ N/m²; using this along with the X-ray diffraction value of $r_0 = 2.82 \times 10^{-10}$ m, we find $n = 1.47$.

Using simple tension in this calculation is not really appropriate, because when a material is stretched in one direction, it will *contract* in the transverse directions. This is the *Poisson effect*, which will be treated in a later module. Our tension-only example does not consider the transverse contraction, and the resulting value of n is too low. A better but slightly more complicated approach is to use hydrostatic

²See the *Module on Crystallographic Notation* for a review of this nomenclature.

compression, which moves all the ions closer to one another. Problem 3 outlines this procedure, which yields values of n in the range of 5–12 as mentioned earlier.

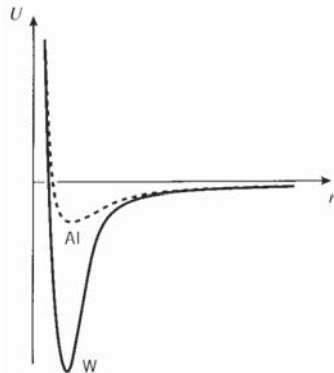


Figure 4: Bond energy functions for aluminum and tungsten.

The stiffnesses of metallic and covalent systems will be calculated differently than the method used above for ionic crystals, but the concept of electrostatic attraction applies to these non-ionic systems as well. As a result, bond energy functions of a qualitatively similar nature result from all these materials. In general, the “tightness” of the bond, and hence the elastic modulus E , is related to the curvature of the bond energy function. Steeper bond functions will also be deeper as a rule, so that within similar classes of materials the modulus tends to correlate with the energy needed to rupture the bonds, for instance by melting. Materials such as tungsten that fill many bonding and few antibonding orbitals have very deep bonding functions³, with correspondingly high stiffnesses and melting temperatures, as illustrated in Fig. 4. This correlation is obvious in Table 1, which lists the values of modulus for a number of metals, along with the values of melting temperature T_m and melting energy ΔH .

Table 1: Modulus and bond strengths for transition metals.

Material	E GPa (Mpsi)	T_m °C	ΔH kJ/mol	α_L $\times 10^{-6}, ^\circ\text{C}^{-1}$
Pb	14 (2)	327	5.4	29
Al	69 (10)	660	10.5	22
Cu	117 (17)	1084	13.5	17
Fe	207 (30)	1538	15.3	12
W	407 (59)	3410	32	4.2

The system will generally have sufficient thermal energy to reside at a level somewhat above the minimum in the bond energy function, and will oscillate between the two positions labeled A and B in Fig. 5, with an average position near r_0 . This simple idealization provides a rationale for why materials expand when the temperature is raised. As the internal energy is increased by the

³A detailed analysis of the cohesive energies of materials is an important topic in solid state physics; see for example F. Seitz, *The Modern Theory of Solids*, McGraw-Hill, 1940.

addition of heat, the system oscillates between the positions labeled A' and B' with an average separation distance r'_0 . Since the curve is anharmonic, the average separation distance is now greater than before, so the material has expanded or stretched. To a reasonable approximation, the relative thermal expansion $\Delta L/L$ is often related linearly to the temperature rise ΔT , and we can write:

$$\boxed{\frac{\Delta L}{L} = \epsilon_T = \alpha_L \Delta T} \quad (7)$$

where ϵ_T is a *thermal strain* and the constant of proportionality α_L is the *coefficient of linear thermal expansion*. The expansion coefficient α_L will tend to correlate with the depth of the energy curve, as is seen in Table 1.

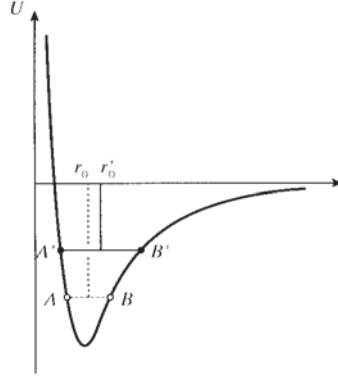


Figure 5: Anharmonicity of the bond energy function.

Example 2

A steel bar of length L and cross-sectional area A is fitted snugly between rigid supports as shown in Fig. 6. We wish to find the compressive stress in the bar when the temperature is raised by an amount ΔT .

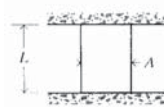


Figure 6: Bar between rigid supports.

If the bar were free to expand, it would increase in length by an amount given by Eqn. 7. Clearly, the rigid supports have to push on the bar – i.e. put in into compression – to suppress this expansion. The magnitude of this thermally induced compressive stress could be found by imagining the material free to expand, then solving $\sigma = E\epsilon_T$ for the stress needed to “push the material back” to its unstrained state. Equivalently, we could simply set the sum of a thermally induced strain and a mechanical strain ϵ_σ to zero:

$$\epsilon = \epsilon_\sigma + \epsilon_T = \frac{\sigma}{E} + \alpha_L \Delta T = 0$$

$$\sigma = -\alpha_L E \Delta T$$

The minus sign in this result reminds us that a negative (compressive) stress is induced by a positive temperature change (temperature rises.)

Example 3

A glass container of stiffness E and thermal expansion coefficient α_L is removed from a hot oven and plunged suddenly into cold water. We know from experience that this “thermal shock” could fracture the glass, and we’d like to see what materials parameters control this phenomenon. The analysis is very similar to that of the previous example.

In the time period just after the cold-water immersion, before significant heat transfer by conduction can take place, the outer surfaces of the glass will be at the temperature of the cold water while the interior is still at the temperature of the oven. The outer surfaces will try to contract, but are kept from doing so by the still-hot interior; this causes a tensile stress to develop on the surface. As before, the stress can be found by setting the total strain to zero:

$$\epsilon = \epsilon_\sigma + \epsilon_T = \frac{\sigma}{E} + \alpha_L \Delta T = 0$$

$$\sigma = -\alpha_L E \Delta T$$

Here the temperature change ΔT is negative if the glass is going from hot to cold, so the stress is positive (tensile). If the glass is not to fracture by thermal shock, this stress must be less than the ultimate tensile strength σ_f ; hence the maximum allowable temperature difference is

$$-\Delta T_{max} = \frac{\sigma_f}{\alpha_L E}$$

To maximize the resistance to thermal shock, the glass should have as low a value of $\alpha_L E$ as possible. “Pyrex” glass was developed specifically for improved thermal shock resistance by using boron rather than soda and lime as process modifiers; this yields a much reduced value of α_L .

Material properties for a number of important structural materials are listed in the *Module on Material Properties*. When the column holding Young’s Modulus is plotted against the column containing the Thermal Expansion Coefficients (using log-log coordinates), the graph shown in Fig. 7 is obtained. Here we see again the general inverse relationship between stiffness and thermal expansion, and the distinctive nature of polymers is apparent as well.

Not all types of materials can be described by these simple bond-energy concepts: *intramolecular* polymer covalent bonds have energies entirely comparable with ionic or metallic bonds, but most common polymers have substantially lower moduli than most metals or ceramics. This is due to the *intermolecular* bonding in polymers being due to secondary bonds which are much weaker than the strong intramolecular covalent bonds. Polymers can also have substantial entropic contributions to their stiffness, as will be described below, and these effects do not necessarily correlate with bond energy functions.

Entropic effects

The internal energy as given by the function $U(r)$ is sufficient to determine the atomic positions in many engineering materials; the material “wants” to minimize its internal energy, and it does this by optimizing the balance of attractive and repulsive electrostatic bonding forces.

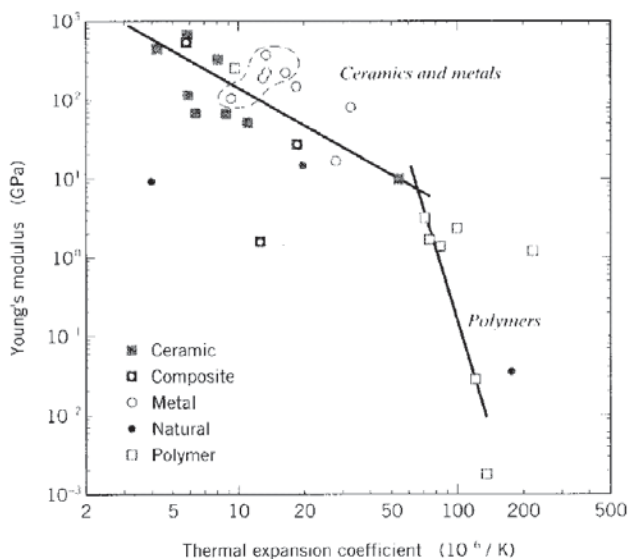


Figure 7: Correlation of stiffness and thermal expansion for materials of various types.

But when the absolute temperature is greater than approximately two-thirds of the melting temperature, there can be sufficient molecular mobility that entropic or disordering effects must be considered as well. This is often the case for polymers even at room temperature, due to their weak intermolecular bonding.

When the temperature is high enough, polymer molecules can be viewed as an interpenetrating mass of (extremely long) wriggling worms, constantly changing their positions by rotation about carbon-carbon single bonds. This wriggling does not require straining the bond lengths or angles, and large changes in position are possible with no change in internal bonding energy.



Figure 8: Conformational change in polymers.

The shape, or “conformation” of a polymer molecule can range from a fully extended chain to a randomly coiled sphere (see Fig. 8). Statistically, the coiled shape is much more likely than the extended one, simply because there are so many ways the chain can be coiled and only one way it can be fully extended. In thermodynamic terms, the entropy of the coiled conformation is very high (many possible “microstates”), and the entropy of the extended conformation is very low (only one possible microstate). If the chain is extended and then released, there will be more wriggling motions tending to the most probable state than to even more highly stretched states; the material would therefore shrink back to its unstretched and highest-entropy state. Equivalently, a person holding the material in the stretched state would feel a tensile force as the material tries to unstretch and is prevented from doing so. These effects are due to entropic

factors, and not internal bond energy.

It is possible for materials to exhibit both internal energy and entropic elasticity. Energy effects dominate in most materials, but rubber is much more dependent on entropic effects. An *ideal rubber* is one in which the response is completely entropic, with the internal energy changes being negligible.

When we stretch a rubber band, the molecules in its interior become extended because they are *crosslinked* by chemical or physical junctions as shown in Fig. 9. Without these links, the molecules could simply slide past one another with little or no uncoiling. “Silly Putty ” is an example of uncrosslinked polymer, and its lack of junction connections cause it to be a viscous fluid rather than a useful elastomer that can bear sustained loads without continuing flow. The crosslinks provide a means by which one molecule can pull on another, and thus establish load transfer within the materials. They also have the effect of limiting how far the rubber can be stretched before breaking, since the extent of the entropic uncoiling is limited by how far the material can extend before pulling up tight against the network of junction points. We will see below that the stiffness of a rubber can be controlled directly by adjusting the crosslink density, and this is an example of process-structure-property control in materials.

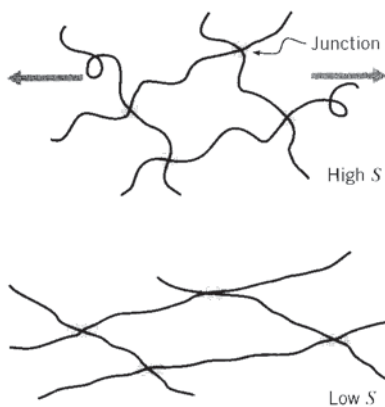


Figure 9: Stretching of crosslinked or entangled polymers.

As the temperature is raised, the Brownian-type wriggling of the polymer is intensified, so that the material seeks more vigorously to assume its random high-entropy state. This means that the force needed to hold a rubber band at fixed elongation *increases* with increasing temperature. Similarly, if the band is stretched by hanging a fixed weight on it, the band will *shrink* as the temperature is raised. In some thermodynamic formalisms it is convenient to model this behavior by letting the coefficient of thermal expansion be a variable parameter, with the ability to become negative for sufficiently large tensile strains. This is a little tricky, however; for instance, the stretched rubber band will contract only along its long axis when the temperature is raised, and will become thicker in the transverse directions. The coefficient of thermal expansion would have to be made not only stretch-dependent but also dependent on direction (“anisotropic”).

Example 4

An interesting demonstration of the unusual thermal response of stretched rubber bands involves replacing the spokes of a bicycle wheel with stretched rubber bands as seen in Fig. 10, then mounting

the wheel so that a heat lamp shines on the bands to the right or left of the hub. As the bands warm up, they contract. This pulls the rim closer to the hub, causing the wheel to become unbalanced. It will then rotate under gravity, causing the warmed bands to move out from under the heat lamp and be replaced other bands. The process continues, and the wheel rotates in a direction opposite to what would be expected were the spokes to expand rather than contract on heating.

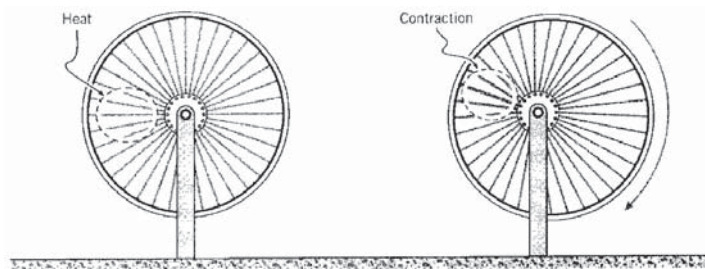


Figure 10: A bicycle wheel with entropic spokes.

The bicycle-wheel trick produces a rather weak response, and it is easy to stop the wheel with only a light touch of the finger. However, the same idea, using very highly stretched urethane bands and employing superheated geothermal steam as a heat source, becomes a viable route for generating mechanical energy.

It is worthwhile to study the response of rubbery materials in some depth, partly because this provides a broader view of the elasticity of materials. But this isn't a purely academic goal. Rubbery materials are being used in increasingly demanding mechanical applications (in addition to tires, which is a very demanding application itself). Elastomeric bearings, vibration-control supports, and biomedical prostheses are but a few examples. We will outline what is known as the "kinetic theory of rubber elasticity," which treats the entropic effect using concepts of statistical thermodynamics. This theory stands as one of the very most successful atomistic theories of mechanical response. It leads to a result of satisfying accuracy without the need for adjustable parameters or other fudge factors.

When pressure-volume changes are not significant, the competition between internal energy and entropy can be expressed by the Helmholtz free energy $A = U - TS$, where T is the temperature and S is the entropy. The system will move toward configurations of lowest free energy, which it can do either by reducing its internal energy or by increasing its entropy. Note that the influence of the entropic term increases explicitly with increasing temperature. With certain thermodynamic limitations in mind (see Prob. 5), the mechanical work $dW = F dL$ done by a force F acting through a differential displacement dL will produce an increase in free energy given by

$$F dL = dW = dU - T dS \quad (8)$$

or

$$F = \frac{dW}{dL} = \left(\frac{\partial U}{\partial L} \right)_{T,V} - T \left(\frac{\partial S}{\partial L} \right)_{T,V} \quad (9)$$

For an ideal rubber, the energy change dU is negligible, so the force is related directly to the temperature and the change in entropy dS produced by the force. To determine the force-deformation relationship, we obviously need to consider how S changes with deformation.

We begin by writing an expression for the conformation, or shape, of the segment of polymer molecule between junction points as a statistical probability distribution. Here the length of the *segment* is the important molecular parameter, not the length of the entire molecule. In the simple form of this theory, which turns out to work quite well, each covalently bonded segment is idealized as a freely-jointed sequence of n rigid links each having length a .

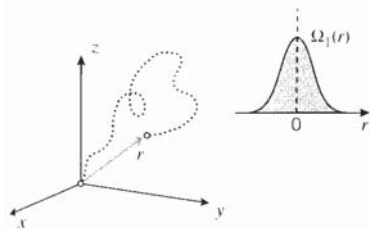


Figure 11: Random-walk model of polymer conformation

A reasonable model for the end-to-end distance of a randomly wriggling segment is that of a “random walk” Gaussian distribution treated in elementary statistics. One end of the chain is visualized at the origin of an xyz coordinate system as shown in Fig. 11, and each successive link in the chain is attached with a random orientation relative to the previous link. (An elaboration of the theory would constrain the orientation so as to maintain the 109° covalent bonding angle.) The probability $\Omega_1(r)$ that the other end of the chain is at a position $r = (x^2 + y^2 + z^2)^{1/2}$ can be shown to be

$$\Omega_1(r) = \frac{\beta^3}{\sqrt{\pi}} \exp(-\beta^2 r^2) = \frac{\beta^3}{\sqrt{\pi}} \exp \left[-\beta^2 (x^2 + y^2 + z^2) \right]$$

The parameter β is a scale factor related to the number of units n in the polymer segment and the bond length a ; specifically it turns out that $\beta = \sqrt{3/2n}/a$. This is the “bell-shaped curve” well known to seasoned test-takers. The most probable end-to-end distance is seen to be zero, which is expected because the chain will end up a given distance to the left (or up, or back) of the origin exactly as often as it ends up the same distance to the right.

When the molecule is now stretched or otherwise deformed, the relative positions of the two ends are changed. Deformation in elastomers is usually described in terms of *extension ratios*, which are the ratios of stretched to original dimensions, L/L_0 . Stretches in the x , y , and z directions are denoted by λ_x , λ_y , and λ_z respectively. The deformation is assumed to be *affine*, i.e. the end-to-end distances of each molecular segment increase by these same ratios. Hence if we continue to view one end of the chain at the origin the other end will have moved to $x_2 = \lambda_x x$, $y_2 = \lambda_y y$, $z_2 = \lambda_z z$. The configurational probability of a segment being found in this stretched state is then

$$\Omega_2 = \frac{\beta^3}{\sqrt{\pi}} \exp \left[-\beta^2 (\lambda_x^2 x^2 + \lambda_y^2 y^2 + \lambda_z^2 z^2) \right]$$

The relative change in probabilities between the perturbed and unperturbed states can now be written as

$$\ln \frac{\Omega_2}{\Omega_1} = -\beta^2 \left[(\lambda_x^2 - 1) x^2 + (\lambda_y^2 - 1) y^2 + (\lambda_z^2 - 1) z^2 \right]$$

Several strategems have been used in the literature to simplify this expression. One simple approach is to let the initial position of the segment end x, y, z be such that $x^2 = y^2 = z^2 = \overline{r_0^2}/3$, where $\overline{r_0^2}$ is the initial mean square end-to-end distance of the segment. (This is not zero, since when squares are taken the negative values no longer cancel the positive ones.) It can also be shown (see Prob. 8) that the distance $\overline{r_0^2}$ is related to the number of bonds n in the segment and the bond length a by $\overline{r_0^2} = na^2$. Making these substitutions and simplifying, we have

$$\ln \frac{\Omega_2}{\Omega_1} = -\frac{1}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3) \quad (10)$$

As is taught in subjects in statistical thermodynamics, changes in configurational probability are related to corresponding changes in thermodynamic entropy by the “Boltzman relation” as

$$\Delta S = k \ln \frac{\Omega_2}{\Omega_1}$$

where $k = 1.38 \times 10^{-23}$ J/K is Boltzman’s constant. Substituting Eqn. 10 in this relation:

$$\Delta S = -\frac{k}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

This is the entropy change for one segment. If there are N chain segments per unit volume, the total entropy change per unit volume ΔS_V is just N times this quantity:

$$\Delta S_V = -\frac{Nk}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3) \quad (11)$$

The associated work (per unit volume) required to change the entropy by this amount is

$$\boxed{\Delta W_V = -T\Delta S_V = +\frac{NkT}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)} \quad (12)$$

The quantity ΔW_V is therefore the strain energy per unit volume contained in an ideal rubber stretched by $\lambda_x, \lambda_y, \lambda_z$.

Example 5

Recent research by Prof. Christine Ortiz has demonstrated that the elasticity of individual polymer chains can be measured using a variety of high-resolution force spectroscopy techniques, such as atomic force microscopy (AFM). At low to moderate extensions, most polymer chains behave as ideal, entropic, random coils; i.e. molecular rubber bands. This is shown in Fig. 12, which displays AFM data (retraction force, F_{chain} , versus chain end-to-end separation distance) for stretching and uncoiling of single polystyrene chains of different lengths. By fitting experimental data with theoretical polymer physics models of freely-jointed chains (red lines in Fig. 12) or worm-like chains, we can estimate the “statistical segment length” or local chain stiffness and use this parameter as a probe of chemical structure and local environmental effects (e.g. electrostatic interactions, solvent quality, etc.). In addition, force spectroscopy can be used to measure noncovalent, physisorption forces of single polymer chains on surfaces and covalent bond strength (chain “fracture”).

To illustrate the use of Eqn. 12 for a simple but useful case, consider a rubber band, initially of length L_0 which is stretched to a new length L . Hence $\lambda = \lambda_x = L/L_0$. To a very good approximation, rubbery materials maintain a constant volume during deformation, and this lets us compute the transverse contractions λ_y and λ_z which accompany the stretch λ_x . An

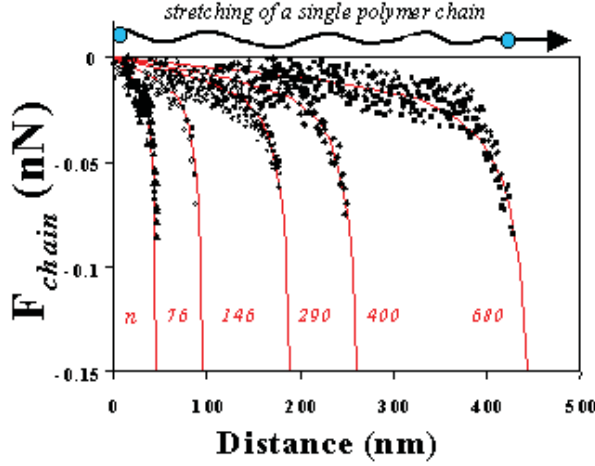


Figure 12: Experimental measurements from numerous force spectroscopy (AFM) experiments of force-elongation response of single polystyrene segments in toluene, compared to the freely-jointed chain model. The statistical segment length is 0.68, and n = number of molecular units in the segment.

expression for the change ΔV in a cubical volume of initial dimensions a_0, b_0, c_0 which is stretched to new dimensions a, b, c is

$$\Delta V = abc - a_0b_0c_0 = (a_0\lambda_x)(b_0\lambda_y)(c_0\lambda_z) - a_0b_0c_0 = a_0b_0c_0(\lambda_x\lambda_y\lambda_z - 1)$$

Setting this to zero gives

$$\lambda_x\lambda_y\lambda_z = 1 \quad (13)$$

Hence the contractions in the y and z directions are

$$\lambda_y^2 = \lambda_z^2 = \frac{1}{\lambda}$$

Using this in Eqn. 12, the force F needed to induce the deformation can be found by differentiating the total strain energy according to Eqn. 9:

$$F = \frac{dW}{dL} = \frac{d(V \Delta W_V)}{L_0 d\lambda} = A_0 \frac{NkT}{2} \left(2\lambda - \frac{2}{\lambda^2} \right)$$

Here $A_0 = V/L_0$ is the original area. Dividing by A_0 to obtain the engineering stress:

$$\boxed{\sigma = NkT \left(\lambda - \frac{1}{\lambda^2} \right)} \quad (14)$$

Clearly, the parameter NkT is related to the stiffness of the rubber, as it gives the stress σ needed to induce a given extension λ . It can be shown (see Prob. 10) that the initial modulus — the slope of the stress-strain curve at the origin — is controlled by the temperature and the crosslink density according to $E = 3NkT$.

Crosslinking in rubber is usually done in the “vulcanizing” process invented by Charles Goodyear in 1839. In this process sulfur abstracts reactive hydrogens adjacent to the double

bonds in the rubber molecule, and forms permanent bridges between adjacent molecules. When crosslinking is done by using approximately 5% sulfur, a conventional rubber is obtained. When the sulfur is increased to $\approx 30\text{--}50\%$, a hard and brittle material named ebonite (or simply “hard rubber”) is produced instead.

The volume density of chain segments N is also the density of junction points. This quantity is related to the specimen density ρ and the molecular weight between crosslinks M_c as $M_c = \rho N_A / N$, where N is the number of crosslinks per unit volume and $N_A = 6.023 \times 10^{23}$ is Avogadro’s Number. When N is expressed in terms of moles per unit volume, we have simply $M_c = \rho / N$ and the quantity NkT in Eqn. 14 is replaced by NRT , where $R = kN_A = 8.314$ J/mol-°K is the Gas Constant.

Example 6

The Young’s modulus of a rubber is measured at $E = 3.5$ MPa for a temperature of $T = 300$ °K. The molar crosslink density is then

$$N = \frac{E}{3RT} = \frac{3.5 \times 10^6 \text{ N/m}^2}{3 \times 8.314 \frac{\text{N}\cdot\text{m}}{\text{mol}\cdot\text{K}} \times 300 \text{ K}} = 468 \text{ mol/m}^3$$

The molecular weight per segment is

$$M_c = \frac{\rho}{N} = \frac{1100 \text{ kg/m}^3}{468 \text{ mol/m}^3} = 2350 \text{ gm/mol}$$

Example 7

A person with more entrepreneurial zeal than caution wishes to start a bungee-jumping company, and naturally wants to know how far the bungee cord will stretch; the clients sometimes complain if the cord fails to stop them before they reach the asphalt. It’s probably easiest to obtain a first estimate from an energy point of view: say the unstretched length of the cord is L_0 , and that this is also the distance the jumper free-falls before the cord begins to stretch. Just as the cord begins to stretch, the the jumper has lost an amount of potential energy wL_0 , where w is the jumper’s weight. The jumper’s velocity at this time could then be calculated from $(mv^2)/2 = wL_0$ if desired, where $m = w/g$ is the jumper’s mass and g is the acceleration of gravity. When the jumper’s velocity has been brought to zero by the cord (assuming the cord doesn’t break first, and the ground doesn’t intervene), this energy will now reside as entropic strain energy within the cord. Using Eqn. 12, we can equate the initial and final energies to obtain

$$wL_0 = \frac{A_0 L_0 \cdot NRT}{2} \left(\lambda^2 + \frac{2}{\lambda} - 3 \right)$$

Here $A_0 L_0$ is the total volume of the cord; the entropic energy per unit volume ΔW_V must be multiplied by the volume to give total energy. Dividing out the initial length L_0 and using $E = 3NRT$, this result can be written in the dimensionless form

$$\frac{w}{A_0 E} = \frac{1}{6} \left(\lambda^2 + \frac{2}{\lambda} - 3 \right)$$

The closed-form solution for λ is messy, but the variable $w/A_0 E$ can easily be plotted versus λ (see Fig. 13.) Note that the length L_0 has canceled from the result, although it is still present implicitly in the extension ratio $\lambda = L/L_0$.

Taking a typical design case for illustration, say the desired extension ratio is taken at $\lambda = 3$ for a rubber cord of initial modulus $E = 100$ psi; this stops the jumper safely above the pavement and is verified

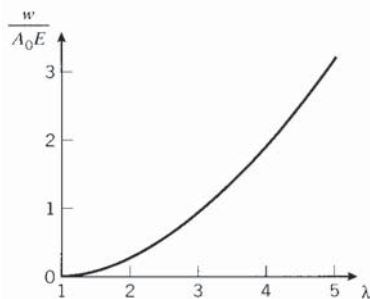


Figure 13: Dimensionless weight versus cord extension.

to be well below the breaking extension of the cord. The value of the parameter w/AE corresponding to $\lambda = 3$ is read from the graph to be 1.11. For a jumper weight of 150 lb, this corresponds to $A = 1.35 \text{ in}^2$, or a cord diameter of 1.31 in.

If ever there was a strong case for field testing, this is it. An analysis such as this is nothing more than a crude starting point, and many tests such as drops with sandbags are obviously called for. Even then, the insurance costs would likely be very substantial.

Note that the stress-strain response for rubber elasticity is nonlinear, and that the stiffness as given by the stress needed to produce a given deformation is predicted to increase with increasing temperature. This is in accord with the concept of more vigorous wriggling with a statistical bias toward the more disordered state. The rubber elasticity equation works well at lower extensions, but tends to deviate from experimental values at high extensions where the segment configurations become nongaussian.

Deviations from Eqn. 14 can also occur due to crystallization at high elongations. (Rubbers are normally noncrystalline, and in fact polymers such as polyethylene that crystallize readily are not elastomeric due to the rigidity imparted by the crystallites.) However, the decreased entropy that accompanies stretching in rubber increases the crystalline melting temperature according to the well-known thermodynamic relation

$$T_m = \frac{\Delta U}{\Delta S} \quad (15)$$

where ΔU and ΔS are the change in internal energy and entropy on crystallization. The quantity ΔS is reduced if stretching has already lowered the entropy, so the crystallization temperature rises. If it rises above room temperature, the rubber develops crystallites that stiffen it considerably and cause further deviation from the rubber elasticity equation. (Since the crystallization is exothermic, the material will also increase in temperature; this can often be sensed by stretching a rubber band and then touching it to the lips.) Strain-induced crystallization also helps inhibit crack growth, and the excellent abrasion resistance of natural rubber is related to the ease with which it crystallizes upon stretching.

Problems

1. Justify the first two terms of the Madelung series given in Eqn. 3.
2. Using Eqn. 6 to write the parameter B in terms of the equilibrium interionic distance r_0 , show that the binding energy of an ionic crystal, per bond pair, can be written as

$$U = -\frac{(n-1)ACe^2}{nr_0}$$

where A is the Madelung constant, C is the appropriate units conversion factor, and e is the ionic charge.

3. Measurements of bulk compressibility are valuable for probing the bond energy function, because unlike simple tension, hydrostatic pressure causes the interionic distance to decrease uniformly. The *modulus of compressibility* K of a solid is the ratio of the pressure p needed to induce a relative change in volume dV/V :

$$K = -\frac{dp}{(dV)/V}$$

The minus sign is needed because positive pressures induce reduced volumes (volume change negative).

- (a) Use the relation $dU = pdV$ for the energy associated with pressure acting through a small volume change to show

$$\frac{K}{V_0} = \left(\frac{d^2U}{dV^2} \right)_{V=V_0}$$

where V_0 is the crystal volume at the equilibrium interionic spacing $r = a_0$.

- (b) The volume of an ionic crystal containing N negative and N positive ions can be written as $V = cNr^3$ where c is a constant dependent on the type of lattice (2 for NaCl). Use this to obtain the relation

$$\frac{K}{V_0} = \left(\frac{d^2U}{dV^2} \right)_{V=V_0} = \frac{1}{9c^2N^2r^2} \cdot \frac{d}{dr} \left(\frac{1}{r^2} \frac{dU}{dr} \right)$$

- (c) Carry out the indicated differentiation of the expression for binding energy to obtain the expression

$$\frac{K}{V_0} = \frac{K}{cNr_0^3} = \frac{N}{9c^2Nr_0^2} \left[\frac{-4ACe}{r_0^5} + \frac{n(n+3)B}{r_0^{n+4}} \right]$$

Then using the expression $B = ACe^2r_0^{n-1}/n$, obtain the formula for n in terms of compressibility:

$$n = 1 + \frac{9cr_0^4K}{ACe^2}$$

4. Complete the spreadsheet below, filling in the values for repulsion exponent n and lattice energy U .

type	r_0 (pm)	K (GPa)	A	n	U (kJ/mol)	U_{expt}
LiF	201.4	6.710e+01	1.750			-1014
NaCl	282.0	2.400e+01	1.750			-764
KBr	329.8	1.480e+01	1.750			-663

The column labeled U_{expt} lists experimentally obtained values of the lattice energy.

5. Given the definition of Helmholtz free energy:

$$A = U - TS$$

along with the first and second laws of thermodynamics:

$$dU = dQ + dW$$

$$dQ = TdS$$

where U is the internal energy, T is the temperature, S is the entropy, Q is the heat and W is the mechanical work, show that the force F required to hold the ends of a tensile specimen a length L apart is related to the Helmholtz energy as

$$F = \left(\frac{\partial A}{\partial L} \right)_{T,V}$$

6. Show that the temperature dependence of the force needed to hold a tensile specimen at fixed length as the temperature is changed (neglecting thermal expansion effects) is related to the dependence of the entropy on extension as

$$\left(\frac{\partial F}{\partial T} \right)_L = - \left(\frac{\partial S}{\partial L} \right)_T$$

7. (a) Show that if an ideal rubber ($dU = 0$) of mass M and specific heat c is extended adiabatically, its temperature will change according to the relation

$$\frac{\partial T}{\partial L} = \frac{-T}{Mc} \left(\frac{\partial S}{\partial L} \right)$$

i.e. if the entropy is reduced upon extension, the temperature will rise. This is known as the *thermoelastic effect*.

- (b) Use this expression to obtain the temperature change dT in terms of an increase $d\lambda$ in the extension ratio as

$$dT = \frac{\sigma}{\rho c} d\lambda$$

where σ is the engineering stress (load divided by original area) and ρ is the mass density.

8. Show that the end-to-end distance r_0 of a chain composed of n freely-jointed links of length a is given by $r_0 = na^2$.

9. Evaluate the temperature rise in a rubber specimen of $\rho = 1100 \text{ kg/m}^3$, $c = 2 \text{ kJ/kg}\cdot\text{K}$, $NkT = 500 \text{ kPa}$, subjected to an axial extension $\lambda = 4$.
10. Show that the initial engineering modulus of a rubber whose stress-strain curve is given by Eqn. 14 is $E = 3NRT$.
11. Calculate the Young's modulus of a rubber of density 1100 gm/mol and whose inter-crosslink segments have a molecular weight of 2500 gm/mol . The temperature is 25°C .
12. Show that in the case of biaxial extension (λ_x and λ_y prescribed), the x -direction stress based on the original cross-sectional dimensions is

$$\sigma_x = NkT \left(\lambda_x - \frac{1}{\lambda_x^2 \lambda_y^2} \right)$$

and based on the deformed dimensions

$$_t\sigma_x = NkT \left(\lambda_x^2 - \frac{1}{\lambda_x^2 \lambda_y^2} \right)$$

where the t subscript indicates a “true” or current stress.

13. Estimate the initial elastic modulus E , at a temperature of 20°C , of an elastomer having a molecular weight of $7,500 \text{ gm/mol}$ between crosslinks and a density of 1.0 gm/cm^3 . What is the percentage change in the modulus if the temperature is raised to 40°C ?
14. Consider a line on a rubber sheet, originally oriented at an angle ϕ_0 from the vertical. When the sheet is stretched in the vertical direction by an amount $\lambda_y = \lambda$, the line rotates to a new inclination angle ϕ' . Show that

$$\tan \phi' = \frac{1}{\lambda^{3/2}} \tan \phi_0$$

15. Before stretching, the molecular segments in a rubber sheet are assumed to be distributed uniformly over all directions, so the the fraction of segments $f(\phi)$ oriented in a particular range of angles $d\phi$ is

$$f(\phi) = \frac{dA}{A} = \frac{2\pi r^2 \sin \phi d\phi}{2\pi r}$$

The *Herrman orientation parameter* is defined in terms of the mean orientation as

$$f = \frac{1}{2} \left(3 \langle \cos^2 \phi' \rangle - 1 \right), \quad \langle \cos^2 \phi' \rangle = \int_0^{\pi/2} \cos^2 \phi' f(\phi) d\phi$$

Using the result of the previous problem, plot the orientation function f as a function of the extension ratio λ .

INTRODUCTION TO COMPOSITE MATERIALS

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March 24, 2000

Introduction

This module introduces basic concepts of stiffness and strength underlying the mechanics of fiber-reinforced advanced composite materials. This aspect of composite materials technology is sometimes terms “micromechanics,” because it deals with the relations between macroscopic engineering properties and the microscopic distribution of the material’s constituents, namely the volume fraction of fiber. This module will deal primarily with unidirectionally-reinforced continuous-fiber composites, and with properties measured along and transverse to the fiber direction.

Materials

The term *composite* could mean almost anything if taken at face value, since all materials are composed of dissimilar subunits if examined at close enough detail. But in modern materials engineering, the term *usually* refers to a “matrix” material that is reinforced with fibers. For instance, the term “FRP” (for Fiber Reinforced Plastic) usually indicates a thermosetting polyester matrix containing glass fibers, and this particular composite has the lion’s share of today’s commercial market. Figure 1 shows a laminate fabricated by “crossplying” unidirectionally-reinforced layers in a 0° - 90° stacking sequence.

Many composites used today are at the leading edge of materials technology, with performance and costs appropriate to ultrademanding applications such as spacecraft. But heterogeneous materials combining the best aspects of dissimilar constituents have been used by nature for millions of years. Ancient society, imitating nature, used this approach as well: the Book of Exodus speaks of using straw to reinforce mud in brickmaking, without which the bricks would have almost no strength.

As seen in Table 1¹, the fibers used in modern composites have strengths and stiffnesses far above those of traditional bulk materials. The high strengths of the glass fibers are due to processing that avoids the internal or surface flaws which normally weaken glass, and the strength and stiffness of the polymeric aramid fiber is a consequence of the nearly perfect alignment of the molecular chains with the fiber axis.

¹F.P. Gerstle, “Composites,” *Encyclopedia of Polymer Science and Engineering*, Wiley, New York, 1991. Here E is Young’s modulus, σ_b is breaking stress, ϵ_b is breaking strain, and ρ is density.

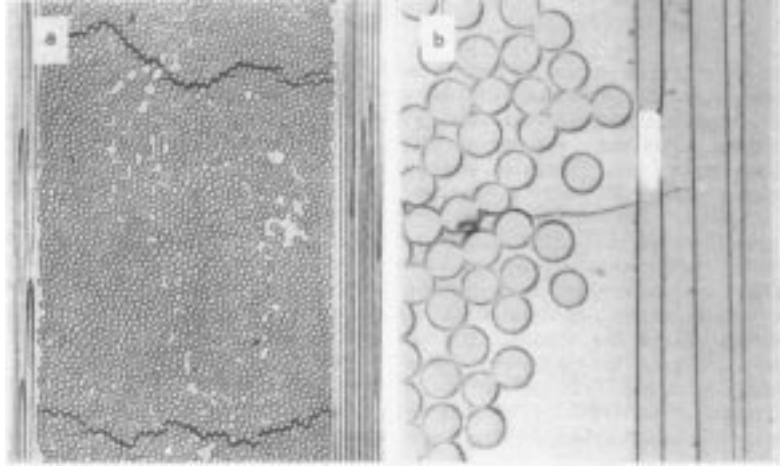


Figure 1: A crossplied FRP laminate, showing nonuniform fiber packing and microcracking (from Harris, 1986).

Table 1: Properties of Composite Reinforcing Fibers.

Material	E (GPa)	σ_b (GPa)	ϵ_b (%)	ρ (Mg/m ³)	E/ρ (MJ/kg)	σ_b/ρ (MJ/kg)	cost (\$/kg)
E-glass	72.4	2.4	2.6	2.54	28.5	0.95	1.1
S-glass	85.5	4.5	2.0	2.49	34.3	1.8	22–33
aramid	124	3.6	2.3	1.45	86	2.5	22–33
boron	400	3.5	1.0	2.45	163	1.43	330–440
HS graphite	253	4.5	1.1	1.80	140	2.5	66–110
HM graphite	520	2.4	0.6	1.85	281	1.3	220–660

Of course, these materials are not generally usable as fibers alone, and typically they are impregnated by a matrix material that acts to transfer loads to the fibers, and also to protect the fibers from abrasion and environmental attack. The matrix dilutes the properties to some degree, but even so very high specific (weight-adjusted) properties are available from these materials. Metal and glass are available as matrix materials, but these are currently very expensive and largely restricted to R&D laboratories. Polymers are much more commonly used, with unsaturated styrene-hardened polyesters having the majority of low-to-medium performance applications and epoxy or more sophisticated thermosets having the higher end of the market. Thermoplastic matrix composites are increasingly attractive materials, with processing difficulties being perhaps their principal limitation.

Stiffness

The fibers may be oriented randomly within the material, but it is also possible to arrange for them to be oriented preferentially in the direction expected to have the highest stresses. Such a material is said to be *anisotropic* (different properties in different directions), and control of the anisotropy is an important means of optimizing the material for specific applications. At a microscopic level, the properties of these composites are determined by the orientation and

distribution of the fibers, as well as by the properties of the fiber and matrix materials. The topic known as composite *micromechanics* is concerned with developing estimates of the overall material properties from these parameters.

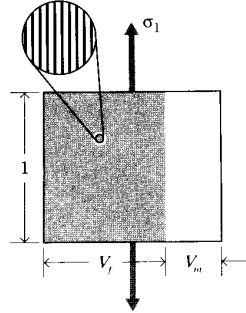


Figure 2: Loading parallel to the fibers.

Consider a typical region of material of unit dimensions, containing a volume fraction V_f of fibers all oriented in a single direction. The matrix volume fraction is then $V_m = 1 - V_f$. This region can be idealized as shown in Fig. 2 by gathering all the fibers together, leaving the matrix to occupy the remaining volume — this is sometimes called the “slab model.” If a stress σ_1 is applied along the fiber direction, the fiber and matrix phases act in *parallel* to support the load. In these parallel connections the strains in each phase must be the same, so the strain ϵ_1 in the fiber direction can be written as:

$$\epsilon_f = \epsilon_m = \epsilon_1$$

The forces in each phase must add to balance the total load on the material. Since the forces in each phase are the phase stresses times the area (here numerically equal to the volume fraction), we have

$$\sigma_1 = \sigma_f V_f + \sigma_m V_m = E_f \epsilon_1 V_f + E_m \epsilon_1 V_m$$

The stiffness in the fiber direction is found by dividing by the strain:

$$E_1 = \frac{\sigma_1}{\epsilon_1} = V_f E_f + V_m E_m \quad (1)$$

This relation is known as a *rule of mixtures* prediction of the overall modulus in terms of the moduli of the constituent phases and their volume fractions.

If the stress is applied in the direction transverse to the fibers as depicted in Fig. 3, the slab model can be applied with the fiber and matrix materials acting *in series*. In this case the stress in the fiber and matrix are equal (an idealization), but the deflections add to give the overall transverse deflection. In this case it can be shown (see Prob. 5)

$$\frac{1}{E_2} = \frac{V_f}{E_f} + \frac{V_m}{E_m} \quad (2)$$

Figure 4 shows the functional form of the parallel (Eqn. 1) and series (Eqn. 2) predictions for the fiber- and transverse-direction moduli.

The prediction of transverse modulus given by the series slab model (Eqn. 2) is considered unreliable, in spite of its occasional agreement with experiment. Among other deficiencies the

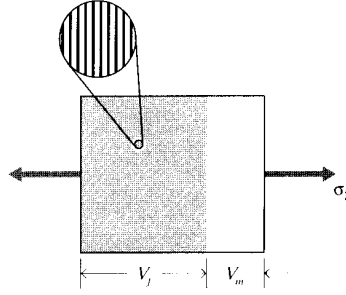


Figure 3: Loading perpendicular to the fibers.

assumption of uniform matrix strain being untenable; both analytical and experimental studies have shown substantial nonuniformity in the matrix strain. Figure 5 shows the photoelastic fringes in the matrix caused by the perturbing effect of the stiffer fibers. (A more complete description of these photoelasticity can be found in the *Module on Experimental Strain Analysis*, but this figure can be interpreted simply by noting that closely-spaced photoelastic fringes are indicative of large strain gradients.

In more complicated composites, for instance those with fibers in more than one direction or those having particulate or other nonfibrous reinforcements, Eqn. 1 provides an *upper bound* to the composite modulus, while Eqn. 2 is a *lower bound* (see Fig. 4). Most practical cases will be somewhere between these two values, and the search for reasonable models for these intermediate cases has occupied considerable attention in the composites research community. Perhaps the most popular model is an empirical one known as the *Halpin-Tsai* equation², which can be written in the form:

$$E = \frac{E_m[E_f + \xi(V_f E_f + V_m E_m)]}{V_f E_m + V_m E_f + \xi E_m} \quad (3)$$

Here ξ is an adjustable parameter that results in series coupling for $\xi = 0$ and parallel averaging for very large ξ .

Strength

Rule of mixtures estimates for strength proceed along lines similar to those for stiffness. For instance, consider a unidirectionally reinforced composite that is strained up to the value at which the fibers begin to break. Denoting this value ϵ_{fb} , the stress transmitted by the composite is given by multiplying the stiffness (Eqn. 1):

$$\sigma_b = \epsilon_{fb} E_1 = V_f \sigma_{fb} + (1 - V_f) \sigma^*$$

The stress σ^* is the stress in the matrix, which is given by $\epsilon_{fb} E_m$. This relation is linear in V_f , rising from σ^* to the fiber breaking strength $\sigma_{fb} = E_f \epsilon_{fb}$. However, this relation is not realistic at low fiber concentration, since the breaking strain of the matrix ϵ_{mb} is usually substantially greater than ϵ_{fb} . If the matrix had *no* fibers in it, it would fail at a stress $\sigma_{mb} = E_m \epsilon_{mb}$. If the fibers were considered to carry no load at all, having broken at $\epsilon = \epsilon_{fb}$ and leaving the matrix

²c.f. J.C.. Halpin and J.L. Kardos, *Polymer Engineering and Science*, Vol. 16, May 1976, pp. 344–352.

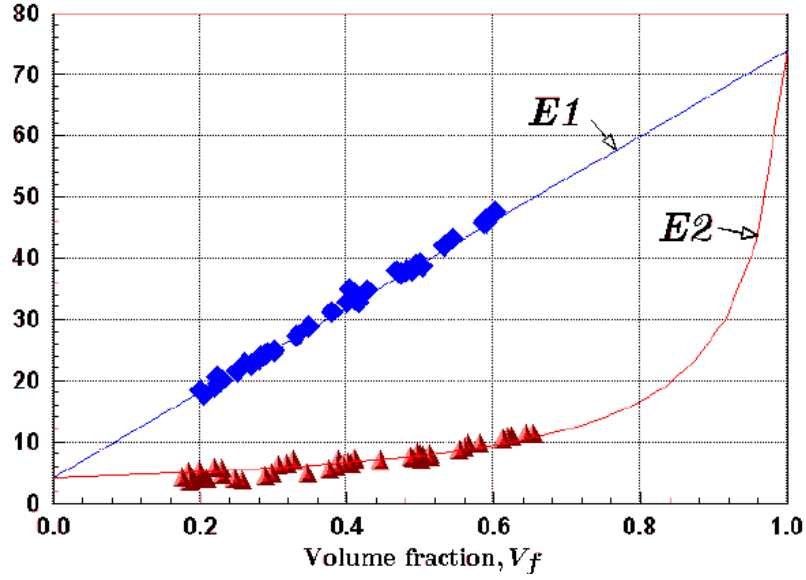


Figure 4: Rule-of-mixtures predictions for longitudinal (E_1) and transverse (E_2) modulus, for glass-polyester composite ($E_f = 73.7$ MPa, $E_m = 4$ GPa). Experimental data taken from Hull (1996).

to carry the remaining load, the strength of the composite would fall off with fiber fraction according to

$$\sigma_b = (1 - V_f)\sigma_{mb}$$

Since the breaking strength actually observed in the composite is the greater of these two expressions, there will be a range of fiber fraction in which the composite is *weakened* by the addition of fibers. These relations are depicted in Fig. 6.

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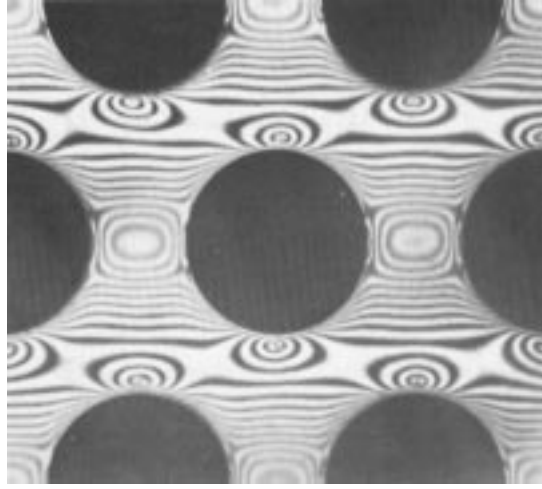


Figure 5: Photoelastic (isochromatic) fringes in a composite model subjected to transverse tension (from Hull, 1996).

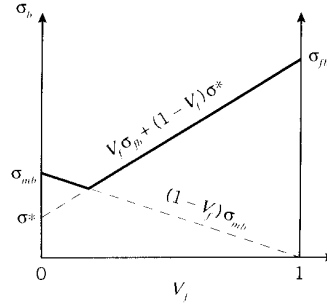


Figure 6: Strength of unidirectional composite in fiber direction.

Problems

1. Compute the longitudinal and transverse stiffness (E_1, E_2) of an S-glass epoxy lamina for a fiber volume fraction $V_f = 0.7$, using the fiber properties from Table 1, and matrix properties from the *Module on Materials Properties*.
2. Plot the longitudinal stiffness E_1 of an E-glass/nylon unidirectionally-reinforced composite, as a function of the volume fraction V_f .
3. Plot the longitudinal tensile strength of a E-glass/epoxy unidirectionally-reinforced composite, as a function of the volume fraction V_f .
4. What is the maximum fiber volume fraction V_f that could be obtained in a unidirectionally reinforced with optimal fiber packing?
5. Using the slab model and assuming uniform strain in the matrix, show the transverse modulus of a unidirectionally-reinforced composite to be

$$\frac{1}{E_2} = \frac{V_f}{E_f} + \frac{V_m}{E_m}$$

or in terms of compliances

$$C_2 = C_f V_f + C_m V_m$$

STRESS-STRAIN CURVES

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August 23, 2001

Introduction

Stress-strain curves are an extremely important graphical measure of a material's mechanical properties, and all students of Mechanics of Materials will encounter them often. However, they are not without some subtlety, especially in the case of ductile materials that can undergo substantial geometrical change during testing. This module will provide an introductory discussion of several points needed to interpret these curves, and in doing so will also provide a preliminary overview of several aspects of a material's mechanical properties. However, this module will not attempt to survey the broad range of stress-strain curves exhibited by modern engineering materials (the atlas by Boyer cited in the References section can be consulted for this). Several of the topics mentioned here — especially yield and fracture — will appear with more detail in later modules.

“Engineering” Stress-Strain Curves

Perhaps the most important test of a material's mechanical response is the tensile test¹, in which one end of a rod or wire specimen is clamped in a loading frame and the other subjected to a controlled displacement δ (see Fig. 1). A transducer connected in series with the specimen provides an electronic reading of the load $P(\delta)$ corresponding to the displacement. Alternatively, modern servo-controlled testing machines permit using load rather than displacement as the controlled variable, in which case the displacement $\delta(P)$ would be monitored as a function of load.

The engineering measures of stress and strain, denoted in this module as σ_e and ϵ_e respectively, are determined from the measured the load and deflection using the original specimen cross-sectional area A_0 and length L_0 as

$$\sigma_e = \frac{P}{A_0}, \quad \epsilon_e = \frac{\delta}{L_0} \quad (1)$$

When the stress σ_e is plotted against the strain ϵ_e , an *engineering stress-strain curve* such as that shown in Fig. 2 is obtained.

¹Stress-strain testing, as well as almost all experimental procedures in mechanics of materials, is detailed by standards-setting organizations, notably the American Society for Testing and Materials (ASTM). Tensile testing of metals is prescribed by ASTM Test E8, plastics by ASTM D638, and composite materials by ASTM D3039.

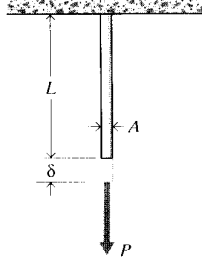


Figure 1: The tension test.

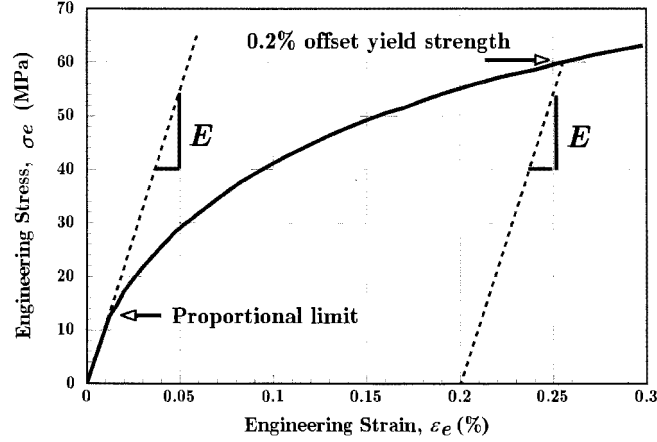


Figure 2: Low-strain region of the engineering stress-strain curve for annealed polycrystalline copper; this curve is typical of that of many ductile metals.

In the early (low strain) portion of the curve, many materials obey Hooke’s law to a reasonable approximation, so that stress is proportional to strain with the constant of proportionality being the modulus of elasticity or Young’s modulus, denoted E :

$$\sigma_e = E\epsilon_e \quad (2)$$

As strain is increased, many materials eventually deviate from this linear proportionality, the point of departure being termed the proportional limit. This nonlinearity is usually associated with stress-induced “plastic” flow in the specimen. Here the material is undergoing a rearrangement of its internal molecular or microscopic structure, in which atoms are being moved to new equilibrium positions. This plasticity requires a mechanism for molecular mobility, which in crystalline materials can arise from dislocation motion (discussed further in a later module.) Materials lacking this mobility, for instance by having internal microstructures that block dislocation motion, are usually brittle rather than ductile. The stress-strain curve for brittle materials are typically linear over their full range of strain, eventually terminating in fracture without appreciable plastic flow.

Note in Fig. 2 that the stress needed to increase the strain beyond the proportional limit in a ductile material continues to rise beyond the proportional limit; the material requires an ever-increasing stress to continue straining, a mechanism termed *strain hardening*.

These microstructural rearrangements associated with plastic flow are usually not reversed

when the load is removed, so the proportional limit is often the same as or at least close to the materials's elastic limit. Elasticity is the property of complete and immediate recovery from an imposed displacement on release of the load, and the elastic limit is the value of stress at which the material experiences a permanent residual strain that is not lost on unloading. The residual strain induced by a given stress can be determined by drawing an unloading line from the highest point reached on the $\sigma_e - \epsilon_e$ curve at that stress back to the strain axis, drawn with a slope equal to that of the initial elastic loading line. This is done because the material unloads elastically, there being no force driving the molecular structure back to its original position.

A closely related term is the yield stress, denoted σ_Y in these modules; this is the stress needed to induce plastic deformation in the specimen. Since it is often difficult to pinpoint the exact stress at which plastic deformation begins, the yield stress is often taken to be the stress needed to induce a specified amount of permanent strain, typically 0.2%. The construction used to find this “offset yield stress” is shown in Fig. 2, in which a line of slope E is drawn from the strain axis at $\epsilon_e = 0.2\%$; this is the unloading line that would result in the specified permanent strain. The stress at the point of intersection with the $\sigma_e - \epsilon_e$ curve is the offset yield stress.

Figure 3 shows the engineering stress-strain curve for copper with an enlarged scale, now showing strains from zero up to specimen fracture. Here it appears that the rate of strain hardening² diminishes up to a point labeled UTS, for Ultimate Tensile Strength (denoted σ_f in these modules). Beyond that point, the material appears to strain soften, so that each increment of additional strain requires a smaller stress.

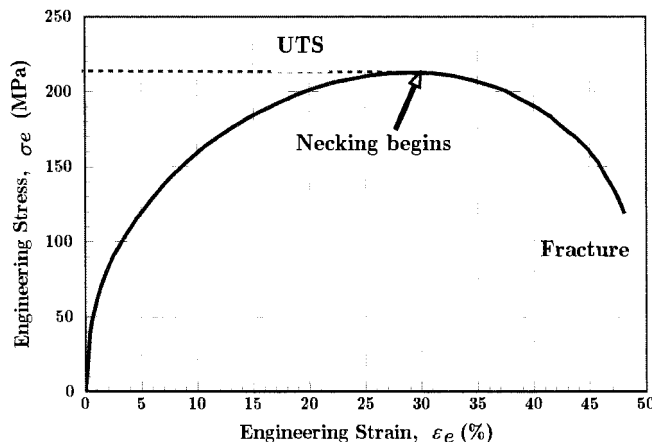


Figure 3: Full engineering stress-strain curve for annealed polycrystalline copper.

The apparent change from strain hardening to strain softening is an artifact of the plotting procedure, however, as is the maximum observed in the curve at the UTS. Beyond the yield point, molecular flow causes a substantial reduction in the specimen cross-sectional area A , so the true stress $\sigma_t = P/A$ actually borne by the material is larger than the engineering stress computed from the original cross-sectional area ($\sigma_e = P/A_0$). The load must equal the true stress times the actual area ($P = \sigma_t A$), and as long as strain hardening can increase σ_t enough to compensate for the reduced area A , the load and therefore the engineering stress will continue to rise as the strain increases. Eventually, however, the decrease in area due to flow becomes larger than the increase in true stress due to strain hardening, and the load begins to fall. This

²The strain hardening rate is the slope of the stress-strain curve, also called the *tangent modulus*.

is a geometrical effect, and if the true stress rather than the engineering stress were plotted no maximum would be observed in the curve.

At the UTS the differential of the load P is zero, giving an analytical relation between the true stress and the area at necking:

$$P = \sigma_t A \rightarrow dP = 0 = \sigma_t dA + A d\sigma_t \rightarrow -\frac{dA}{A} = \frac{d\sigma_t}{\sigma_t} \quad (3)$$

The last expression states that the load and therefore the engineering stress will reach a maximum as a function of strain when the fractional decrease in area becomes equal to the fractional increase in true stress.

Even though the UTS is perhaps the materials property most commonly reported in tensile tests, it is not a direct measure of the material due to the influence of geometry as discussed above, and should be used with caution. The yield stress σ_Y is usually preferred to the UTS in designing with ductile metals, although the UTS is a valid design criterion for brittle materials that do not exhibit these flow-induced reductions in cross-sectional area.

The true stress is not quite uniform throughout the specimen, and there will always be some location - perhaps a nick or some other defect at the surface - where the local stress is maximum. Once the maximum in the engineering curve has been reached, the localized flow at this site cannot be compensated by further strain hardening, so the area there is reduced further. This increases the local stress even more, which accelerates the flow further. This localized and increasing flow soon leads to a “neck” in the gage length of the specimen such as that seen in Fig. 4.



Figure 4: Necking in a tensile specimen.

Until the neck forms, the deformation is essentially uniform throughout the specimen, but after necking all subsequent deformation takes place in the neck. The neck becomes smaller and smaller, local true stress increasing all the time, until the specimen fails. This will be the failure mode for most ductile metals. As the neck shrinks, the nonuniform geometry there alters the uniaxial stress state to a complex one involving shear components as well as normal stresses. The specimen often fails finally with a “cup and cone” geometry as seen in Fig. 5, in which the outer regions fail in shear and the interior in tension. When the specimen fractures, the engineering strain at break — denoted ϵ_f — will include the deformation in the necked region and the unnecked region together. Since the true strain in the neck is larger than that in the unnecked material, the value of ϵ_f will depend on the fraction of the gage length that has necked. Therefore, ϵ_f is a function of the specimen geometry as well as the material, and thus is only a

crude measure of material ductility.



Figure 5: Cup-and-cone fracture in a ductile metal.

Figure 6 shows the engineering stress-strain curve for a semicrystalline thermoplastic. The response of this material is similar to that of copper seen in Fig. 3, in that it shows a proportional limit followed by a maximum in the curve at which necking takes place. (It is common to term this maximum as the yield stress in plastics, although plastic flow has actually begun at earlier strains.)

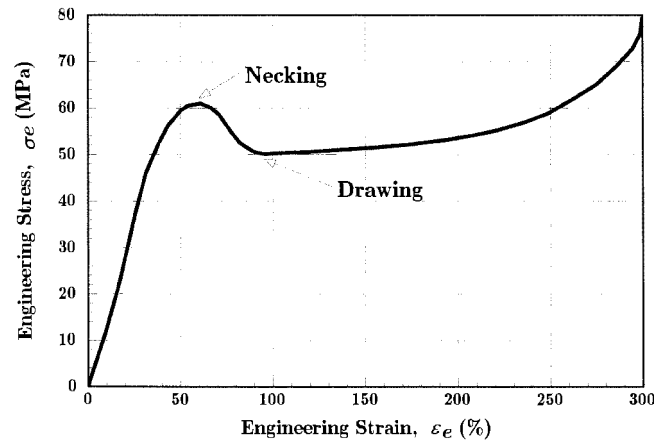


Figure 6: Stress-strain curve for polyamide (nylon) thermoplastic.

The polymer, however, differs dramatically from copper in that the neck does not continue shrinking until the specimen fails. Rather, the material in the neck stretches only to a “natural draw ratio” which is a function of temperature and specimen processing, beyond which the material in the neck stops stretching and new material at the neck shoulders necks down. The neck then propagates until it spans the full gage length of the specimen, a process called *drawing*. This process can be observed without the need for a testing machine, by stretching a polyethylene “six-pack holder,” as seen in Fig. 7.

Not all polymers are able to sustain this drawing process. As will be discussed in the next section, it occurs when the necking process produces a strengthened microstructure whose breaking load is greater than that needed to induce necking in the untransformed material just outside the neck.



Figure 7: Necking and drawing in a 6-pack holder.

“True” Stress-Strain Curves

As discussed in the previous section, the engineering stress-strain curve must be interpreted with caution beyond the elastic limit, since the specimen dimensions experience substantial change from their original values. Using the true stress $\sigma_t = P/A$ rather than the engineering stress $\sigma_e = P/A_0$ can give a more direct measure of the material’s response in the plastic flow range. A measure of strain often used in conjunction with the true stress takes the increment of strain to be the incremental increase in displacement dL divided by the current length L :

$$d\epsilon_t = \frac{dL}{L} \rightarrow \epsilon_t = \int_{L_0}^L \frac{1}{L} dL = \ln \frac{L}{L_0} \quad (4)$$

This is called the “true” or “logarithmic” strain.

During yield and the plastic-flow regime following yield, the material flows with negligible change in volume; increases in length are offset by decreases in cross-sectional area. Prior to necking, when the strain is still uniform along the specimen length, this volume constraint can be written:

$$dV = 0 \rightarrow AL = A_0L_0 \rightarrow \frac{L}{L_0} = \frac{A}{A_0} \quad (5)$$

The ratio L/L_0 is the *extension ratio*, denoted as λ . Using these relations, it is easy to develop relations between true and engineering measures of tensile stress and strain (see Prob. 2):

$$\sigma_t = \sigma_e (1 + \epsilon_e) = \sigma_e \lambda, \quad \epsilon_t = \ln(1 + \epsilon_e) = \ln \lambda \quad (6)$$

These equations can be used to derive the true stress-strain curve from the engineering curve, up to the strain at which necking begins. Figure 8 is a replot of Fig. 3, with the true stress-strain curve computed by this procedure added for comparison.

Beyond necking, the strain is nonuniform in the gage length and to compute the true stress-strain curve for greater engineering strains would not be meaningful. However, a complete true stress-strain curve could be drawn if the neck area were monitored throughout the tensile test, since for logarithmic strain we have

$$\frac{L}{L_0} = \frac{A}{A_0} \rightarrow \epsilon_t = \ln \frac{L}{L_0} = \ln \frac{A}{A_0} \quad (7)$$

Ductile metals often have true stress-strain relations that can be described by a simple power-law relation of the form:

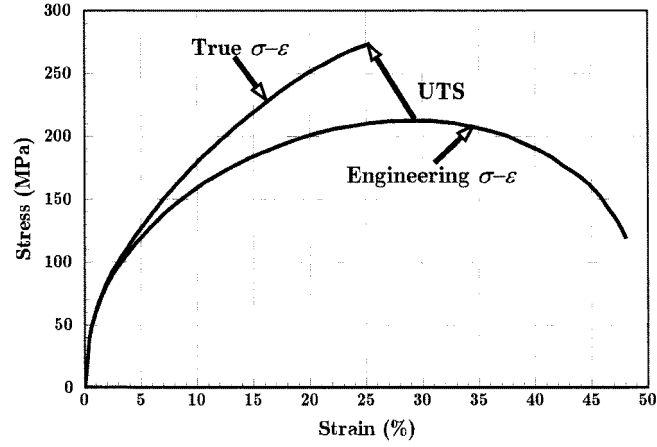


Figure 8: Comparison of engineering and true stress-strain curves for copper. An arrow indicates the position on the “true” curve of the UTS on the engineering curve.

$$\sigma_t = A\epsilon_t^n \rightarrow \log \sigma_t = \log A + n \log \epsilon_t \quad (8)$$

Figure 9 is a log-log plot of the true stress-strain data³ for copper from Fig. 8 that demonstrates this relation. Here the parameter $n = 0.474$ is called the *strain hardening parameter*, useful as a measure of the resistance to necking. Ductile metals at room temperature usually exhibit values of n from 0.02 to 0.5.

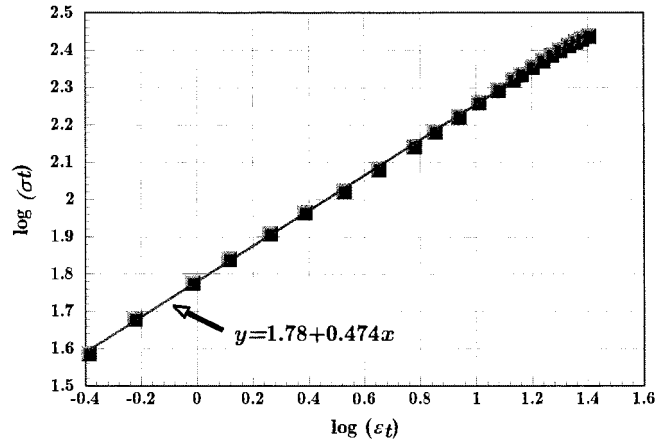


Figure 9: Power-law representation of the plastic stress-strain relation for copper.

A graphical method known as the “Considère construction” uses a form of the true stress-strain curve to quantify the differences in necking and drawing from material to material. This method replots the tensile stress-strain curve with true stress σ_t as the ordinate and extension ratio $\lambda = L/L_0$ as the abscissa. From Eqn. 6, the engineering stress σ_e corresponding to any

³Here percent strain was used for ϵ_t ; this produces the same value for n but a different A than if full rather than percentage values were used.

value of true stress σ_t is slope of a secant line drawn from origin ($\lambda = 0$, not $\lambda = 1$) to intersect the $\sigma_t - \lambda$ curve at σ_t .

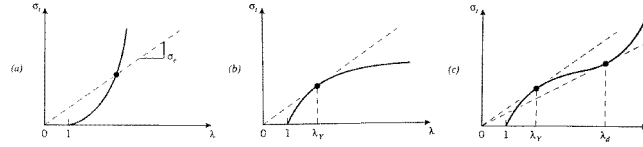


Figure 10: Considère construction. (a) True stress-strain curve with no tangents - no necking or drawing. (b) One tangent - necking but not drawing. (c) Two tangents - necking and drawing.

Among the many possible shapes the true stress-strain curves could assume, let us consider the concave up, concave down, and sigmoidal shapes shown in Fig. 10. These differ in the number of tangent points that can be found for the secant line, and produce the following yield characteristics:

- (a) *No tangents:* Here the curve is always concave upward as in part (a) of Fig. 10, so the slope of the secant line rises continuously. Therefore the engineering stress rises as well, without showing a yield drop. Eventually fracture intercedes, so a true stress-strain curve of this shape identifies a material that fractures before it yields.
- (b) *One tangent:* The curve is concave downward as in part (b) of Fig. 10, so a secant line reaches a tangent point at $\lambda = \lambda_Y$. The slope of the secant line, and therefore the engineering stress as well, begins to fall at this point. This is then the yield stress σ_Y seen as a maximum in stress on a conventional stress-strain curve, and λ_Y is the extension ratio at yield. The yielding process begins at some adventitious location in the gage length of the specimen, and continues at that location rather than being initiated elsewhere because the secant modulus has been reduced at the first location. The specimen is now flowing at a single location with decreasing resistance, leading eventually to failure. Ductile metals such as aluminum fail in this way, showing a marked reduction in cross sectional area at the position of yield and eventual fracture.
- (c) *Two tangents:* For sigmoidal stress-strain curves as in part (c) of Fig. 10, the engineering stress begins to fall at an extension ration λ_Y , but then rises again at λ_d . As in the previous one-tangent case, material begins to yield at a single position when $\lambda = \lambda_Y$, producing a neck that in turn implies a nonuniform distribution of strain along the gage length. Material at the neck location then stretches to λ_d , after which the engineering stress there would have to rise to stretch it further. But this stress is greater than that needed to stretch material at the edge of the neck from λ_Y to λ_d , so material already in the neck stops stretching and the neck propagates outward from the initial yield location. Only material within the neck shoulders is being stretched during propagation, with material inside the necked-down region holding constant at λ_d , the material's "natural draw ratio," and material outside holding at λ_Y . When all the material has been drawn into the necked region, the stress begins to rise uniformly in the specimen until eventually fracture occurs.

The increase in strain hardening rate needed to sustain the drawing process in semicrystalline polymers arises from a dramatic transformation in the material's microstructure. These materials are initially "spherulitic," containing flat lamellar crystalline plates, perhaps 10 nm

thick, arranged radially outward in a spherical domain. As the induced strain increases, these spherulites are first deformed in the straining direction. As the strain increases further, the spherulites are broken apart and the lamellar fragments rearranged with a dominantly axial molecular orientation to become what is known as the fibrillar microstructure. With the strong covalent bonds now dominantly lined up in the load-bearing direction, the material exhibits markedly greater strengths and stiffnesses — by perhaps an order of magnitude — than in the original material. This structure requires a much higher strain hardening rate for increased strain, causing the upturn and second tangent in the true stress-strain curve.

Strain energy

The area under the $\sigma_e - \epsilon_e$ curve up to a given value of strain is the total mechanical energy per unit volume consumed by the material in straining it to that value. This is easily shown as follows:

$$U^* = \frac{1}{V} \int P dL = \int_0^L \frac{P}{A_0} \frac{dL}{L_0} = \int_0^\epsilon \sigma d\epsilon \quad (9)$$

In the absence of molecular slip and other mechanisms for energy dissipation, this mechanical energy is stored reversibly within the material as strain energy. When the stresses are low enough that the material remains in the elastic range, the strain energy is just the triangular area in Fig. 11:

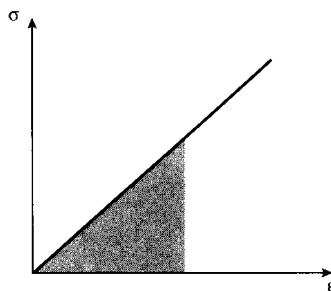


Figure 11: Strain energy = area under stress-strain curve.

Note that the strain energy increases quadratically with the stress or strain; i.e. that as the strain increases the energy stored by a given increment of additional strain grows as the square of the strain. This has important consequences, one example being that an archery bow cannot be simply a curved piece of wood to work well. A real bow is initially straight, then bent when it is strung; this stores substantial strain energy in it. When it is bent further on drawing the arrow back, the energy available to throw the arrow is very much greater than if the bow were simply carved in a curved shape without actually bending it.

Figure 12 shows schematically the amount of strain energy available for two equal increments of strain $\Delta\epsilon$, applied at different levels of existing strain.

The area up to the yield point is termed the *modulus of resilience*, and the total area up to fracture is termed the *modulus of toughness*; these are shown in Fig. 13. The term “modulus” is used because the units of strain energy per unit volume are $\text{N}\cdot\text{m}/\text{m}^3$ or N/m^2 , which are the same as stress or modulus of elasticity. The term “resilience” alludes to the concept that up to the point of yielding, the material is unaffected by the applied stress and upon unloading

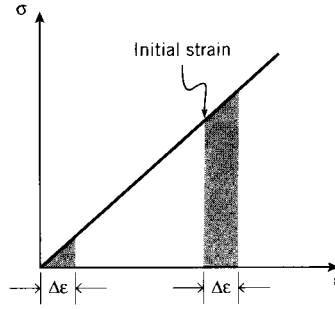


Figure 12: Energy associated with increments of strain

Table 1: Energy absorption of various materials.

Material	Maximum Strain, %	Maximum Stress, MPa	Modulus of Toughness, MJ/m ³	Density kg/m ³	Max. Energy J/kg
Ancient Iron	0.03	70	0.01	7,800	1.3
Modern spring steel	0.3	700	1.0	7,800	130
Yew wood	0.3	120	0.5	600	900
Tendon	8.0	70	2.8	1,100	2,500
Rubber	300	7	10.0	1,200	8,000

will return to its original shape. But when the strain exceeds the yield point, the material is deformed irreversibly, so that some residual strain will persist even after unloading. The modulus of resilience is then the quantity of energy the material can absorb without suffering damage. Similarly, the modulus of toughness is the energy needed to completely fracture the material. Materials showing good impact resistance are generally those with high moduli of toughness.

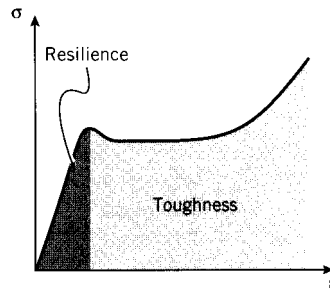


Figure 13: Moduli of resilience and toughness.

Table 1⁴ lists energy absorption values for a number of common materials. Note that natural and polymeric materials can provide extremely high energy absorption per unit weight.

During loading, the area under the stress-strain curve is the strain energy per unit volume absorbed by the material. Conversely, the area under the unloading curve is the energy released by the material. In the elastic range, these areas are equal and no net energy is absorbed. But

⁴J.E. Gordon, *Structures, or Why Things Don't Fall Down*, Plenum Press, New York, 1978.

if the material is loaded into the plastic range as shown in Fig. 14, the energy absorbed exceeds the energy released and the difference is dissipated as heat.

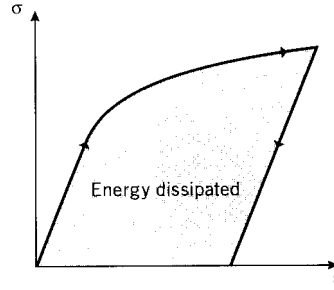


Figure 14: Energy loss = area under stress-strain loop.

Compression

The above discussion is concerned primarily with simple tension, i.e. uniaxial loading that increases the interatomic spacing. However, as long as the loads are sufficiently small (stresses less than the proportional limit), in many materials the relations outlined above apply equally well if loads are placed so as to put the specimen in compression rather than tension. The expression for deformation and a given load $\delta = PL/AE$ applies just as in tension, with negative values for δ and P indicating compression. Further, the modulus E is the same in tension and compression to a good approximation, and the stress-strain curve simply extends as a straight line into the third quadrant as shown in Fig. 15.

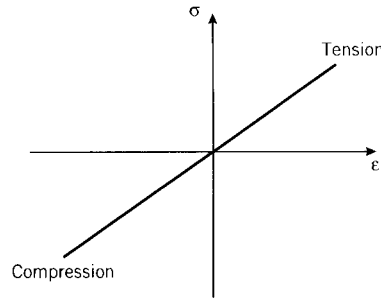


Figure 15: Stress-strain curve in tension and compression.

There are some practical difficulties in performing stress-strain tests in compression. If excessively large loads are mistakenly applied in a tensile test, perhaps by wrong settings on the testing machine, the specimen simply breaks and the test must be repeated with a new specimen. But in compression, a mistake can easily damage the load cell or other sensitive components, since even after specimen failure the loads are not necessarily relieved.

Specimens loaded cyclically so as to alternate between tension and compression can exhibit hysteresis loops if the loads are high enough to induce plastic flow (stresses above the yield stress). The enclosed area in the loop seen in Fig. 16 is the strain energy per unit volume released as heat in each loading cycle. This is the well-known tendency of a wire that is being

bent back and forth to become quite hot at the region of plastic bending. The temperature of the specimen will rise according to the magnitude of this internal heat generation and the rate at which the heat can be removed by conduction within the material and convection from the specimen surface.

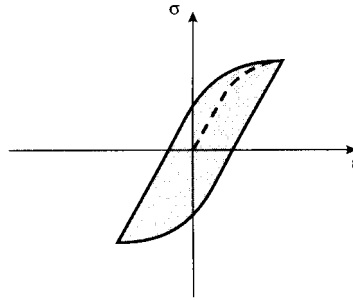


Figure 16: Hysteresis loop.

Specimen failure by cracking is inhibited in compression, since cracks will be closed up rather than opened by the stress state. A number of important materials are much stronger in compression than in tension for this reason. Concrete, for example, has good compressive strength and so finds extensive use in construction in which the dominant stresses are compressive. But it has essentially no strength in tension, as cracks in sidewalks and building foundations attest: tensile stresses appear as these structures settle, and cracks begin at very low tensile strain in unreinforced concrete.

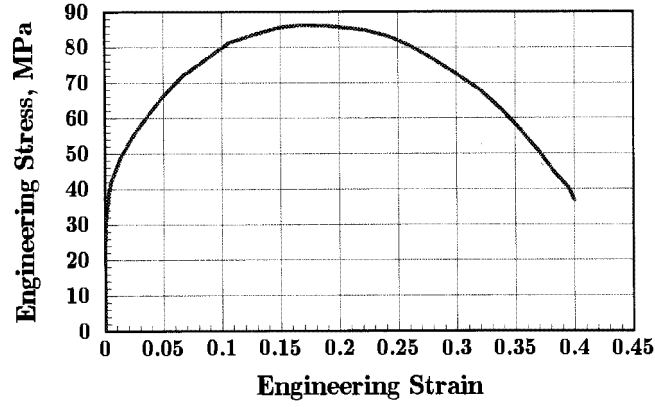
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1. Boyer, H.F., *Atlas of Stress-Strain Curves*, ASM International, Metals Park, Ohio, 1987.
2. Courtney, T.H., *Mechanical Behavior of Materials*, McGraw-Hill, New York, 1990.
3. Hayden, H.W., W.G. Moffatt and J. Wulff, *The Structure and Properties of Materials: Vol. III Mechanical Behavior*, Wiley, New York, 1965.

Problems

1. The figure below shows the engineering stress-strain curve for pure polycrystalline aluminum; the numerical data for this figure are in the file `aluminum.txt`, which can be imported into a spreadsheet or other analysis software. For this material, determine (a) Young's modulus, (b) the 0.2% offset yield strength, (c) the Ultimate Tensile Strength (UTS), (d) the modulus of resilience, and (e) the modulus of toughness.
2. Develop the relations given in Eqn. 6:

$$\sigma_t = \sigma_e (1 + \epsilon_e) = \sigma_e \lambda, \quad \epsilon_t = \ln(1 + \epsilon_e) = \ln \lambda$$



Prob. 1

3. Using the relations of Eqn. 6, plot the true stress-strain curve for aluminum (using data from Prob.1) up to the strain of neck formation.
4. Replot the the results of the previous problem using log-log axes as in Fig. 9 to determine the parameters A and n in Eqn. 8 for aluminum.
5. Using Eqn. 8 with parameters $A = 800$ MPa, $n = 0.2$, plot the engineering stress-strain curve up to a strain of $\epsilon_e = 0.4$. Does the material neck? Explain why the curve is or is not valid at strains beyond necking.
6. Using the parameters of the previous problem, use the condition $(d\sigma_e/d\epsilon_e)_{\text{neck}} = 0$ to show that the engineering strain at necking is $\epsilon_{e,\text{neck}} = 0.221$.
7. Use a Considère construction (plot σ_t vs. λ , as in Fig. 10) to verify the result of the previous problem.
8. Elastomers (rubber) have stress-strain relations of the form

$$\sigma_e = \frac{E}{3} \left(\lambda - \frac{1}{\lambda^2} \right),$$

where E is the initial modulus. Use the Considère construction to show whether this material will neck, or draw.

9. Show that a power-law material (one obeying Eqn. 8) necks when the true strain ϵ_t becomes equal to the strain-hardening exponent n .
10. Show that the UTS (engineering stress at incipient necking) for a power-law material (Eqn. 8) is

$$\sigma_f = \frac{An^n}{e^n}$$

11. Show that the strain energy $U = \int \sigma d\epsilon$ can be computed using either engineering or true values of stress and strain, with equal result.

12. Show that the strain energy needed to neck a power-law material (Eqn.8) is

$$U = \frac{An^{n+1}}{n+1}$$

The Kinematic Equations

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September 19, 2000

Introduction

The *kinematic* or *strain-displacement* equations describe how the strains – the stretching and distortion – within a loaded body relate to the body's displacements. The displacement components in the x , y , and z directions are denoted by the vector $\mathbf{u} \equiv u_i \equiv (u, v, w)$, and are functions of position within the body: $\mathbf{u} = \mathbf{u}(x, y, z)$. If all points within the material experience the same displacement ($\mathbf{u} = \text{constant}$), the structure moves as a rigid body, but does not stretch or deform internally. For stretching to occur, points within the body must experience *different* displacements.

Infinitesimal strain

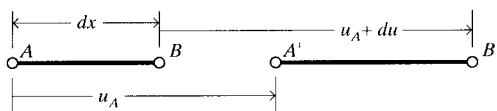


Figure 1: Incremental deformation.

Consider two points A and B separated initially by a small distance dx as shown in Fig. 1, and experiencing motion in the x direction. If the displacement at point A is u_A , the displacement at B can be expressed by a Taylor's series expansion of $u(x)$ around the point $x = A$:

$$u_B = u_A + du = u_A + \frac{\partial u}{\partial x} dx$$

where here the expansion has been truncated after the second term. The *differential* motion δ between the two points is then

$$\delta = u_B - u_A = \left(u_A + \frac{\partial u}{\partial x} dx \right) - u_A = \frac{\partial u}{\partial x} dx$$

In our concept of stretching as being the differential displacement per unit length, the x component of strain is then

$$\boxed{\epsilon_x = \frac{\delta}{dx} = \frac{\partial u}{\partial x}} \quad (1)$$

Hence the strain is a *displacement gradient*. Applying similar reasoning to differential motion in the y direction, the y -component of strain is the gradient of the vertical displacement v with respect to y :

$$\boxed{\epsilon_y = \frac{\partial v}{\partial y}} \quad (2)$$

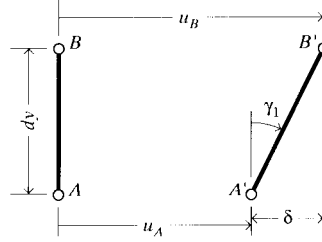


Figure 2: Shearing distortion.

The distortion of the material, which can be described as the change in originally right angles, is the sum of the tilts imparted to vertical and horizontal lines. As shown in Fig. 2, the tilt of an originally vertical line is the relative horizontal displacement of two nearby points along the line:

$$\delta = u_B - u_A = \left(u_A + \frac{\partial u}{\partial y} dy \right) - u_A = \frac{\partial u}{\partial y} dy$$

The change in angle is then

$$\gamma_1 \approx \tan \gamma_1 = \frac{\delta}{dy} = \frac{\partial u}{\partial y}$$

Similarly (see Fig. 3), the tilt γ_2 of an originally horizontal line is the gradient of v with respect to x . The shear strain in the xy plane is then

$$\boxed{\gamma_{xy} = \gamma_1 + \gamma_2 = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}} \quad (3)$$

This notation, using ϵ for normal strain and γ for shearing strain, is sometimes known as the “classical” description of strain.

Matrix Formulation

The “indicial notation” described in the Module on Matrix and Index Notation provides a concise method of writing out all the components of three-dimensional states of strain:

$$\boxed{\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \equiv \frac{1}{2} (u_{i,j} + u_{j,i})} \quad (4)$$

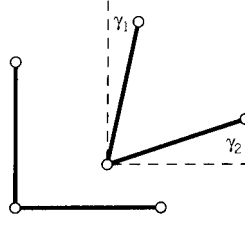


Figure 3: Shearing strain.

where the comma denotes differentiation with respect to the following spatial variable. This double-subscript index notation leads naturally to a matrix arrangement of the strain components, in which the i - j component of the strain becomes the matrix element in the i^{th} row and the j^{th} column:

$$\epsilon_{ij} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) & \frac{\partial w}{\partial z} \end{bmatrix} \quad (5)$$

Note that the strain matrix is symmetric, i.e. $\epsilon_{ij} = \epsilon_{ji}$. This symmetry means that there are six rather than nine independent strains, as might be expected in a 3×3 matrix. Also note that the indicial description of strain yields the same result for the normal components as in the classical description: $\epsilon_{11} = \epsilon_x$. However, the indicial components of shear strain are half their classical counterparts: $\epsilon_{12} = \gamma_{xy}/2$.

In still another useful notational scheme, the classical strain-displacement equations can be written out in a vertical list, similar to a vector:

$$\begin{pmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{pmatrix} = \begin{pmatrix} \partial u / \partial x \\ \partial v / \partial y \\ \partial w / \partial z \\ \partial v / \partial z + \partial w / \partial y \\ \partial u / \partial z + \partial w / \partial x \\ \partial u / \partial y + \partial v / \partial x \end{pmatrix}$$

This vector-like arrangement of the strain components is for convenience only, and is sometimes called a *pseudovector*. Strain is actually a *second-rank tensor*, like stress or moment of inertia, and has mathematical properties very different than those of vectors. The ordering of the elements in the pseudovector form is arbitrary, but it is conventional to list them as we have here by moving down the diagonal of the strain matrix of Eqn. 5 from upper left to lower right, then move up the third column, and finally move one column to the left on the first row; this gives the ordering 1,1; 2,2; 3,3; 2,3; 1,3; 1,2.

Following the rules of matrix multiplication, the strain pseudovector can also be written in

terms of the displacement vector as

$$\begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \partial/\partial x & 0 & 0 \\ 0 & \partial/\partial y & 0 \\ 0 & 0 & \partial/\partial z \\ 0 & \partial/\partial z & \partial/\partial y \\ \partial/\partial z & 0 & \partial/\partial x \\ \partial/\partial y & \partial/\partial x & 0 \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix} \quad (6)$$

The matrix in brackets above, whose elements are differential operators, can be abbreviated as **L**:

$$\mathbf{L} = \begin{bmatrix} \partial/\partial x & 0 & 0 \\ 0 & \partial/\partial y & 0 \\ 0 & 0 & \partial/\partial z \\ 0 & \partial/\partial z & \partial/\partial y \\ \partial/\partial z & 0 & \partial/\partial x \\ \partial/\partial y & \partial/\partial x & 0 \end{bmatrix} \quad (7)$$

The strain-displacement equations can then be written in the concise “pseudovector-matrix” form:

$$\boxed{\boldsymbol{\epsilon} = \mathbf{L}\mathbf{u}} \quad (8)$$

Equations such as this must be used in a well-defined context, as they apply only when the somewhat arbitrary pseudovector listing of the strain components is used.

Volumetric strain

Since the normal strain is just the change in length per unit of original length, the new length L' after straining is found as

$$\epsilon = \frac{L' - L_0}{L_0} \Rightarrow L' = (1 + \epsilon)L_0 \quad (9)$$

If a cubical volume element, originally of dimension abc , is subjected to normal strains in all three directions, the change in the element’s volume is

$$\begin{aligned} \frac{\Delta V}{V} &= \frac{a'b'c' - abc}{abc} = \frac{a(1 + \epsilon_x) b(1 + \epsilon_y) c(1 + \epsilon_z) - abc}{abc} \\ &= (1 + \epsilon_x)(1 + \epsilon_y)(1 + \epsilon_z) - 1 \approx \epsilon_x + \epsilon_y + \epsilon_z \end{aligned} \quad (10)$$

where products of strains are neglected in comparison with individual values. The volumetric strain is therefore the sum of the normal strains, i.e. the sum of the diagonal elements in the strain matrix (this is also called the *trace* of the matrix, or $\text{Tr}[\boldsymbol{\epsilon}]$). In index notation, this can be written simply

$$\frac{\Delta V}{V} = \epsilon_{kk}$$

This is known as the volumetric, or “dilatational” component of the strain.

Example 1

To illustrate how volumetric strain is calculated, consider a thin sheet of steel subjected to strains in its plane given by $\epsilon_x = 3$, $\epsilon_y = -4$, and $\gamma_{xy} = 6$ (all in $\mu\text{in/in}$). The sheet is not in plane strain, since it can undergo a Poisson strain in the z direction given by $\epsilon_z = -\nu(\epsilon_x + \epsilon_y) = -0.3(3 - 4) = 0.3$. The total state of strain can therefore be written as the matrix

$$[\epsilon] = \begin{bmatrix} 3 & 6 & 0 \\ 6 & -4 & 0 \\ 0 & 0 & 0.3 \end{bmatrix} \times 10^{-6}$$

where the brackets on the $[\epsilon]$ symbol emphasize that the matrix rather than pseudovector form of the strain is being used. The volumetric strain is:

$$\frac{\Delta V}{V} = (3 - 4 + 0.3) \times 10^{-6} = -0.7 \times 10^{-6}$$

Engineers often refer to “microinches” of strain; they really mean microinches per inch. In the case of volumetric strain, the corresponding (but awkward) unit would be micro-cubic-inches per cubic inch.

Finite strain

The infinitesimal strain-displacement relations given by Eqns. 3.1–3.3 are used in the vast majority of mechanical analyses, but they do not describe stretching accurately when the displacement gradients become large. This often occurs when polymers (especially elastomers) are being considered. Large strains also occur during deformation processing operations, such as stamping of steel automotive body panels. The kinematics of large displacement or strain can be complicated and subtle, but the following section will outline a simple description of *Lagrangian* finite strain to illustrate some of the concepts involved.

Consider two orthogonal lines OB and OA as shown in Fig. 4, originally of length dx and dy , along the x - y axes, where for convenience we set $dx = dy = 1$. After strain, the endpoints of these lines move to new positions $A_1O_1B_1$ as shown. We will describe these new positions using the coordinate scheme of the original x - y axes, although we could also allow the new positions to define a new set of axes. In following the motion of the lines with respect to the original positions, we are using the so-called *Lagrangian* viewpoint. We could alternately have used the final positions as our reference; this is the *Eulerian* view often used in fluid mechanics.

After straining, the distance dx becomes

$$(dx)' = \left(1 + \frac{\partial u}{\partial x}\right) dx$$

Using our earlier “small” thinking, the x -direction strain would be just $\partial u/\partial x$. But when the strains become larger, we must also consider that the upward motion of point B_1 relative to O_1 , that is $\partial v/\partial x$, also helps stretch the line OB . Considering both these effects, the Pythagorean theorem gives the new length O_1B_1 as

$$O_1B_1 = \sqrt{\left(1 + \frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2}$$

We now define our Lagrangian strain as

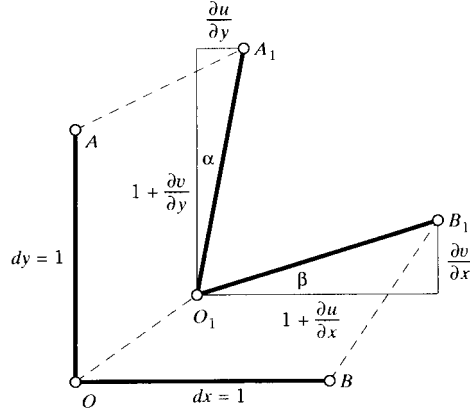


Figure 4: Finite displacements.

$$\begin{aligned}\epsilon_x &= \frac{O_1B_1 - OB}{OB} = O_1B_1 - 1 \\ &= \sqrt{1 + 2\frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2} - 1\end{aligned}$$

Using the series expansion $\sqrt{1+x} = 1 + x/2 + x^2/8 + \dots$ and neglecting terms beyond first order, this becomes

$$\begin{aligned}\epsilon_x &\approx \left\{ 1 + \frac{1}{2} \left[2\frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right] \right\} - 1 \\ &= \frac{\partial u}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right]\end{aligned}\tag{11}$$

Similarly, we can show

$$\epsilon_y = \frac{\partial v}{\partial y} + \frac{1}{2} \left[\left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 \right]\tag{12}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial x}\tag{13}$$

When the strains are sufficiently small that the quadratic terms are negligible compared with the linear ones, these reduce to the infinitesimal-strain expressions shown earlier.

Example 2

The displacement function $u(x)$ for a tensile specimen of uniform cross section and length L , fixed at one end and subjected to a displacement δ at the other, is just the linear relation

$$u(x) = \left(\frac{x}{L}\right) \delta$$

The Lagrangian strain is then given by Eqn. 11 as

$$\epsilon_x = \frac{\delta}{L} + \frac{1}{2} \left(\frac{\delta}{L} \right)^2$$

The first term is the familiar small-strain expression, with the second nonlinear term becoming more important as δ becomes larger. When $\delta = L$, i.e. the conventional strain is 100%, there is a 50% difference between the conventional and Lagrangian strain measures.

The Lagrangian strain components can be generalized using index notation as

$$\epsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i} + u_{r,i} u_{r,j}).$$

A pseudovector form is also convenient occasionally:

$$\begin{aligned} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix} &= \begin{Bmatrix} u_{,x} \\ v_{,y} \\ u_{,y} + v_{,x} \end{Bmatrix} + \frac{1}{2} \begin{bmatrix} u_{,x} & v_{,x} & 0 & 0 \\ 0 & 0 & u_{,y} & v_{,y} \\ u_{,y} & v_{,y} & u_{,x} & v_{,x} \end{bmatrix} \begin{Bmatrix} u_{,x} \\ v_{,x} \\ u_{,y} \\ v_{,y} \end{Bmatrix} \\ &= \left(\begin{bmatrix} \partial/\partial x & 0 \\ 0 & \partial/\partial y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} u_{,x} & v_{,x} & 0 & 0 \\ 0 & 0 & u_{,y} & v_{,y} \\ u_{,y} & v_{,y} & u_{,x} & v_{,x} \end{bmatrix} \begin{bmatrix} \partial/\partial x & 0 \\ 0 & \partial/\partial x \\ \partial/\partial y & 0 \\ 0 & \partial/\partial y \end{bmatrix} \right) \begin{Bmatrix} u \\ v \end{Bmatrix} \end{aligned}$$

which can be abbreviated

$$\boldsymbol{\epsilon} = [\mathbf{L} + \mathbf{A}(\mathbf{u})] \mathbf{u} \quad (14)$$

The matrix $\mathbf{A}(\mathbf{u})$ contains the nonlinear effect of large strain, and becomes negligible when strains are small.

Problems

1. Write out the abbreviated strain-displacement equation $\boldsymbol{\epsilon} = \mathbf{L}\mathbf{u}$ (Eqn. 8) for two dimensions.
2. Write out the components of the Lagrangian strain tensor in three dimensions:

$$\epsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i} + u_{r,i} u_{r,j})$$

3. Show that for small strains the fractional volume change is the trace of the infinitesimal strain tensor:

$$\frac{\Delta V}{V} \equiv \epsilon_{kk} = \epsilon_x + \epsilon_y + \epsilon_z$$

4. When the material is incompressible, show the extension ratios are related by

$$\lambda_x \lambda_y \lambda_z = 1$$

5. Show that the kinematic (strain-displacement) relations in for polar coordinates can be written

$$\epsilon_r = \frac{\partial u_r}{\partial r}$$

$$\epsilon_\theta = \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}$$

$$\gamma_{r\theta} = \frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}$$

The Equilibrium Equations

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September 26, 2000

Introduction

The kinematic relations described in Module 8 are purely geometric, and do not involve considerations of material behavior. The equilibrium relations to be discussed in this module have this same independence from the material. They are simply Newton's law of motion, stating that in the absence of acceleration all of the forces acting on a body (or a piece of it) must balance. This allows us to state how the stress within a body, but evaluated just below the surface, is related to the external force applied to the surface. It also governs how the stress varies from position to position within the body.

Cauchy stress

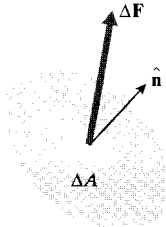


Figure 1: Traction vector.

In earlier modules, we expressed the normal stress as force per unit area acting perpendicularly to a selected area, and a shear stress was a force per unit area acting transversely to the area. To generalize this concept, consider the situation depicted in Fig. 1, in which a *traction vector* $\mathbf{T}$ acts on an arbitrary plane within or on the external boundary of the body, and at an arbitrary direction with respect to the orientation of the plane. The traction is a simple force vector having magnitude and direction, but its magnitude is expressed in terms of force per unit of area:

$$\boxed{\mathbf{T} = \lim_{\Delta A \rightarrow 0} \left(\frac{\Delta \mathbf{F}}{\Delta A} \right)} \quad (1)$$

where ΔA is the magnitude of the area on which $\Delta \mathbf{F}$ acts. The *Cauchy*¹ stresses, which are a generalization of our earlier definitions of stress, are the forces per unit area acting on the Cartesian x , y , and z planes to balance the traction. In two dimensions this balance can be written by drawing a simple free body diagram with the traction vector acting on an area of arbitrary size A (Fig. 2), remembering to obtain the forces by multiplying by the appropriate area.

$$\sigma_x(A \cos \theta) + \tau_{xy}(A \sin \theta) = T_x A$$

$$\tau_{xy}(A \cos \theta) + \sigma_y(A \sin \theta) = T_y A$$

Canceling the factor A , this can be written in matrix form as

$$\begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{Bmatrix} \cos \theta \\ \sin \theta \end{Bmatrix} = \begin{Bmatrix} T_x \\ T_y \end{Bmatrix} \quad (2)$$

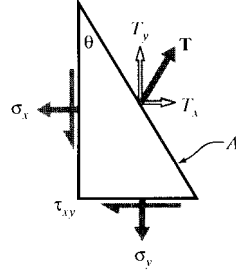


Figure 2: Cauchy stress.

Example 1

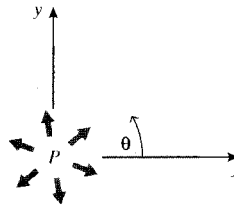


Figure 3: Constant pressure on internal circular boundary.

Consider a circular cavity containing an internal pressure p . The components of the traction vector are then $T_x = -p \cos \theta$, $T_y = -p \sin \theta$. The Cartesian Cauchy stresses in the material at the boundary must then satisfy the relations

$$\sigma_x \cos \theta + \tau_{xy} \sin \theta = -p \cos \theta$$

¹Baron Augustin-Louis Cauchy (1789–1857) was a prolific French engineer and mathematician.

$$\tau_{xy} \cos \theta + \sigma_y \sin \theta = -p \sin \theta$$

At $\theta = 0$, $\sigma_x = -p$, $\sigma_y = \tau_{xy} = 0$; at $\theta = \pi/2$, $\sigma_y = -p$, $\sigma_x = \tau_{xy} = 0$. The shear stress τ_{xy} vanishes for $\theta = 0$ or $\pi/2$; in Module 10 it will be seen that the normal stresses σ_x and σ_y are therefore *principal* stresses at those points.

The vector $(\cos \theta, \sin \theta)$ on the left hand side of Eqn. 2 is also the vector $\hat{\mathbf{n}}$ of direction cosines of the normal to the plane on which the traction acts, and serves to define the orientation of this plane. This matrix equation, which is sometimes called *Cauchy's relation*, can be abbreviated as

$$[\boldsymbol{\sigma}] \hat{\mathbf{n}} = \mathbf{T} \quad (3)$$

The brackets here serve as a reminder that the stress is being written as the square matrix of Eqn. 2 rather than in pseudovector form. This relation serves to define the stress concept as an entity that relates the traction (a vector) acting on an arbitrary surface to the orientation of the surface (another vector). The stress is therefore of a higher degree of abstraction than a vector, and is technically a *second-rank tensor*. The difference between vectors (first-rank tensors) and second-rank tensors shows up in how they *transform* with respect to coordinate rotations, which is treated in Module 10. As illustrated by the previous example, Cauchy's relation serves both to define the stress and to compute its magnitude at boundaries where the tractions are known.

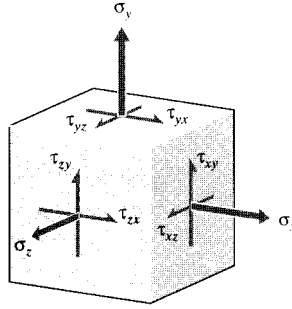


Figure 4: Cartesian Cauchy stress components in three dimensions.

In three dimensions, the matrix form of the stress state shown in Fig. 4 is the symmetric 3×3 array obtained by an obvious extension of the one in Eqn. 2:

$$[\boldsymbol{\sigma}] = \sigma_{ij} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \quad (4)$$

The element in the i^{th} row and the j^{th} column of this matrix is the stress on the i^{th} face in the j^{th} direction. Moment equilibrium requires that the stress matrix be symmetric, so the order of subscripts of the off-diagonal shearing stresses is immaterial.

Differential governing equations

Determining the variation of the stress components as functions of position within the interior of a body is obviously a principal goal in stress analysis. This is a type of *boundary value problem* often encountered in the theory of differential equations, in which the *gradients* of the variables, rather than the explicit variables themselves, are specified. In the case of stress, the gradients are governed by conditions of static equilibrium: the stresses cannot change arbitrarily between two points A and B , or the material between those two points may not be in equilibrium.

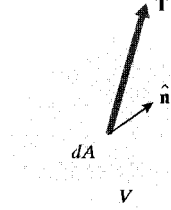


Figure 5: Traction vector $\mathbf{T}$ acting on differential area dA with direction cosines $\hat{\mathbf{n}}$.

To develop this idea formally, we require that the integrated value of the surface traction $\mathbf{T}$ over the surface A of an arbitrary volume element dV within the material (see Fig. 5) must sum to zero in order to maintain static equilibrium :

$$0 = \int_A \mathbf{T} dA = \int_A [\boldsymbol{\sigma}] \hat{\mathbf{n}} dA$$

Here we assume the lack of gravitational, centripetal, or other “body” forces acting on material within the volume. The surface integral in this relation can be converted to a volume integral by Gauss’ divergence theorem²:

$$\int_V \nabla [\boldsymbol{\sigma}] dV = 0$$

Since the volume V is arbitrary, this requires that the integrand be zero:

$$\nabla [\boldsymbol{\sigma}] = 0 \tag{5}$$

For Cartesian problems in three dimensions, this expands to:

$$\boxed{\begin{aligned} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} &= 0 \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} &= 0 \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} &= 0 \end{aligned}} \tag{6}$$

Using index notation, these can be written:

$$\sigma_{ij,j} = 0 \tag{7}$$

²Gauss’ Theorem states that $\int_A X \hat{\mathbf{n}} dA = \int_V \nabla X dV$ where X is a scalar, vector, or tensor quantity.

Or in pseudovector-matrix form, we can write

$$\begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 & 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & 0 & \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \\ 0 & 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} \quad (8)$$

Noting that the differential operator matrix in the brackets is just the transform of the one that appeared in Eqn. 7 of Module 8, we can write this as:

$$\mathbf{L}^T \boldsymbol{\sigma} = \mathbf{0} \quad (9)$$

Example 2

It isn't hard to come up with functions of stress that satisfy the equilibrium equations; any constant will do, since the stress gradients will then be identically zero. The catch is that they must satisfy the boundary conditions as well, and this complicates things considerably. Later modules will outline several approaches to solving the equations directly, but in some simple cases a solution can be seen by inspection.

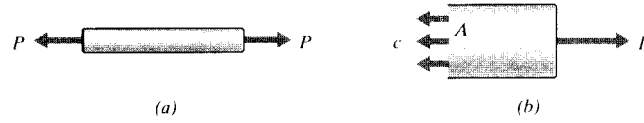


Figure 6: A tensile specimen.

Consider a tensile specimen subjected to a load P as shown in Fig. 6. A trial solution that certainly satisfies the equilibrium equations is

$$[\boldsymbol{\sigma}] = \begin{bmatrix} c & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

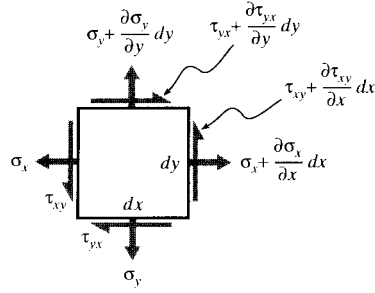
where c is a constant we must choose so as to satisfy the boundary conditions. To maintain horizontal equilibrium in the free-body diagram of Fig. 6(b), it is immediately obvious that $cA = P$, or $\sigma_x = c = P/A$. This familiar relation was used in Module 1 to define the stress, but we see here that it can be viewed as a consequence of equilibrium considerations rather than a basic definition.

Problems

1. Determine whether the following stress state satisfies equilibrium:

$$[\boldsymbol{\sigma}] = \begin{bmatrix} 2x^3y^2 & -2x^2y^3 \\ -2x^2y^3 & xy^4 \end{bmatrix}$$

2. Develop the two-dimensional form of the Cartesian equilibrium equations by drawing a free-body diagram of an infinitesimal section:



Prob. 2

3. Use the free body diagram of the previous problem to show that $\tau_{xy} = \tau_{yx}$.
4. Use a free-body diagram approach to show that in polar coordinates the equilibrium equations are

$$\frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

$$\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + 2 \frac{\tau_{r\theta}}{r} = 0$$

5. Develop the above equations for equilibrium in polar coordinates by transforming the Cartesian equations using

$$x = r \cos \theta$$

$$y = r \sin \theta$$

6. The *Airy stress function* $\phi(x, y)$ is defined such that the Cartesian Cauchy stresses are

$$\sigma_x = \frac{\partial^2 \phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y}$$

Show that the stresses obtained from this procedure satisfy the equilibrium equations.

Transformation of Stresses and Strains

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May 14, 2001

Introduction

One of the most common problems in mechanics of materials involves *transformation of axes*. For instance, we may know the stresses acting on xy planes, but are really more interested in the stresses acting on planes oriented at, say, 30° to the x axis as seen in Fig. 1, perhaps because these are close-packed atomic planes on which sliding is prone to occur, or is the angle at which two pieces of lumber are glued together in a “scarf” joint. We seek a means to transform the stresses to these new $x'y'$ planes.

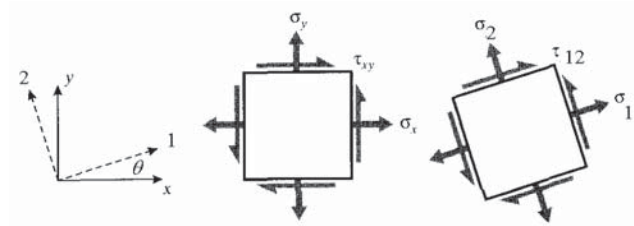


Figure 1: Rotation of axes in two dimensions.

These transformations are vital in analyses of stress and strain, both because they are needed to compute critical values of these entities and also because the tensorial nature of stress and strain is most clearly seen in their transformation properties. Other entities, such as moment of inertia and curvature, also transform in a manner similar to stress and strain. All of these are *second-rank tensors*, an important concept that will be outlined later in this module.

Direct approach

The rules for stress transformations can be developed directly from considerations of static equilibrium. For illustration, consider the case of uniaxial tension shown in Fig. 2 in which all stresses other than σ_y are zero. A free body diagram is then constructed in which the specimen is “cut” along the inclined plane on which the stresses, labeled $\sigma_{y'}$ and $\tau_{x'y'}$, are desired. The key here is to note that the area on which these transformed stresses act is different than the area normal to the y axis, so that both the areas and the forces acting on them need to be “transformed.” Balancing forces in the y' direction (the direction normal to the inclined plane):

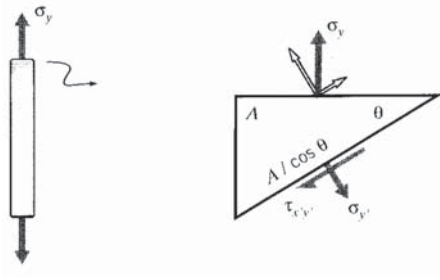


Figure 2: An inclined plane in a tensile specimen.

$$\begin{aligned}
 (\sigma_y A) \cos \theta &= \sigma_{y'} \left(\frac{A}{\cos \theta} \right) \\
 \sigma_{y'} &= \sigma_y \cos^2 \theta
 \end{aligned} \tag{1}$$

Similarly, a force balance in the tangential direction gives

$$\tau_{x'y'} = \sigma_y \sin \theta \cos \theta \tag{2}$$

Example 1

Consider a unidirectionally reinforced composite ply with strengths $\hat{\sigma}_1$ in the fiber direction, $\hat{\sigma}_2$ in the transverse direction, and $\hat{\tau}_{12}$ in shear. As the angle θ between the fiber direction and an applied tensile stress σ_y is increased, the stress in the fiber direction will decrease according to Eqn. 1. If the ply were to fail by fiber fracture alone, the stress $\sigma_{y,b}$ needed to cause failure would *increase* with misalignment according to $\sigma_{y,b} = \hat{\sigma}_1 / \cos^2 \theta$.

However, the shear stresses as given by Eqn. 2 *increase* with θ , so the σ_y stress needed for shear failure drops. The strength $\sigma_{y,b}$ is the smaller of the stresses needed to cause fiber-direction or shear failure, so the strength becomes limited by shear after only a few degrees of misalignment. In fact, a 15° off-axis tensile specimen has been proposed as a means of measuring intralaminar shear strength. When the orientation angle approaches 90° , failure is dominated by the transverse strength. The experimental data shown in Fig. 3 are for glass-epoxy composites¹, which show good but not exact agreement with these simple expressions.

A similar approach, but generalized to include stresses σ_x and τ_{xy} on the original xy planes as shown in Fig. 4 (see Prob. 2) gives:

$$\begin{aligned}
 \sigma_{x'} &= \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta \\
 \sigma_{y'} &= \sigma_x \sin^2 \theta + \sigma_y \cos^2 \theta - 2\tau_{xy} \sin \theta \cos \theta \\
 \tau_{x'y'} &= (\sigma_y - \sigma_x) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)
 \end{aligned} \tag{3}$$

These relations can be written in pseudovector-matrix form as

$$\begin{Bmatrix} \sigma_{x'} \\ \sigma_{y'} \\ \tau_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} \tag{4}$$

¹R.M. Jones, *Mechanics of Composite Materials*, McGraw-Hill, 1975.

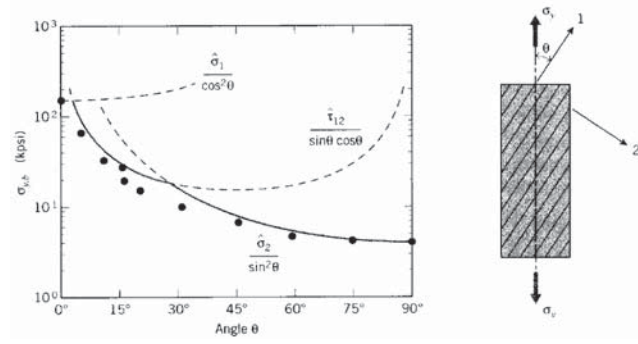


Figure 3: Stress applied at an angle to the fibers in a one-dimensional ply.

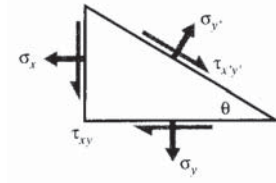


Figure 4: Stresses on inclined plane.

where $c = \cos \theta$ and $s = \sin \theta$. This can be abbreviated as

$$\boldsymbol{\sigma}' = \mathbf{A}\boldsymbol{\sigma} \quad (5)$$

where $\mathbf{A}$ is the transformation matrix in brackets above. This expression would be valid for three dimensional as well as two dimensional stress states, although the particular form of $\mathbf{A}$ given in Eqn. 4 is valid in two dimensions only (plane stress), and for Cartesian coordinates.

Using either mathematical or geometric arguments (see Probs. 3 and 4), it can be shown that the components of infinitesimal strain transform by *almost* the same relations:

$$\begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \frac{1}{2}\gamma_{x'y'} \end{Bmatrix} = \mathbf{A} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix} \quad (6)$$

The factor of 1/2 on the shear components arises from the classical definition of shear strain, which is twice the tensorial shear strain. This introduces some awkwardness into the transformation relations, some of which can be reduced by defining the *Reuter's matrix* as

$$[\mathbf{R}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{or} \quad [\mathbf{R}]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \quad (7)$$

We can now write:

$$\begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \gamma_{x'y'} \end{Bmatrix} = \mathbf{R} \begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \frac{1}{2}\gamma_{x'y'} \end{Bmatrix} = \mathbf{R}\mathbf{A} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix} = \mathbf{R}\mathbf{A}\mathbf{R}^{-1} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

Or

$$\epsilon' = \mathbf{R} \mathbf{A} \mathbf{R}^{-1} \epsilon \quad (8)$$

As can be verified by expanding this relation, the transformation equations for strain can also be obtained from the stress transformation equations (e.g. Eqn. 3) by replacing σ with ϵ and τ with $\gamma/2$:

$$\begin{aligned} \epsilon_{x'} &= \epsilon_x \cos^2 \theta + \epsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cos \theta \\ \epsilon_{y'} &= \epsilon_x \sin^2 \theta + \epsilon_y \cos^2 \theta - \gamma_{xy} \sin \theta \cos \theta \\ \gamma_{x'y'} &= 2(\epsilon_y - \epsilon_x) \sin \theta \cos \theta + \gamma_{xy} (\cos^2 \theta - \sin^2 \theta) \end{aligned} \quad (9)$$

Example 2

Consider the biaxial strain state

$$\epsilon = \begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \gamma_{x'y'} \end{Bmatrix} = \begin{Bmatrix} 0.01 \\ -0.01 \\ 0 \end{Bmatrix}$$

The state of strain ϵ' referred to axes rotated by $\theta = 45^\circ$ from the x - y axes can be computed by matrix multiplication as:

$$\mathbf{A} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 1.0 \\ 0.5 & 0.5 & -1.0 \\ -0.5 & 0.5 & 0.0 \end{bmatrix}$$

Then

$$\begin{aligned} \epsilon' &= \mathbf{R} \mathbf{A} \mathbf{R}^{-1} \epsilon \\ &= \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 2.0 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 & 1.0 \\ 0.5 & 0.5 & -1.0 \\ -0.5 & 0.5 & 0.0 \end{bmatrix} \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 0.5 \end{bmatrix} = \begin{Bmatrix} 0.00 \\ 0.00 \\ -0.02 \end{Bmatrix} \end{aligned}$$

Obviously, the matrix multiplication method is tedious unless matrix-handling software is available, in which case it becomes very convenient.

Mohr's circle

Everyday experience with such commonplace occurrences as pushing objects at an angle gives us all a certain intuitive sense of how vector transformations work. Second-rank tensor transformations seem more abstract at first, and a device to help visualize them is of great value. As it happens, the transformation equations have a famous (among engineers) graphical interpretation known as *Mohr's circle*². The Mohr procedure is justified mathematically by using the trigonometric double-angle relations to show that Eqns. 3 have a circular representation (see Prob. 5), but it can probably best be learned simply by memorizing the following recipe³:

²Presented in 1900 by the German engineer Otto Mohr (1835–1918).

³An interactive web demonstration of Mohr's circle construction is available at <http://web.mit.edu/course/3/3.11/www/java/mohr.html>.

1. Draw the stress square, noting the values on the x and y faces; Fig. 5(a) shows a hypothetical case for illustration. *For the purpose of Mohr's circle only*, regard a shear stress acting in a clockwise-rotation sense as being positive, and counter-clockwise as negative. The shear stresses on the x and y faces must then have opposite signs. The normal stresses are positive in tension and negative in compression, as usual.

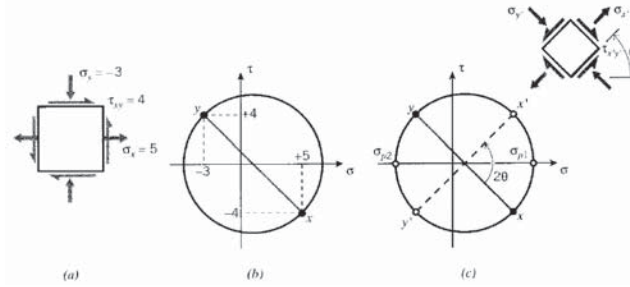


Figure 5: Stress square (a) and Mohr's circle (b) for $\sigma_x = +5$, $\sigma_y = -3$, $\tau_{xy} = +4$. (c) Stress state on inclined plane.

2. Construct a graph with τ as the ordinate (y axis) and σ as abscissa, and plot the stresses on the x and y faces of the stress square as two points on this graph. Since the shear stresses on these two faces are the negative of one another, one of these points will be above the σ -axis exactly as far as the other is below. It is helpful to label the two points as x and y .
3. Connect these two points with a straight line. It will cross the σ axis at the line's midpoint. This point will be at $(\sigma_x + \sigma_y)/2$, which in our illustration is $[5 + (-3)]/2 = 1$.
4. Place the point of a compass at the line's midpoint, and set the pencil at the end of the line. Draw a circle with the line as a diameter. The completed circle for our illustrative stress state is shown in Fig. 5(b).
5. To determine the stresses on a stress square that has been rotated through an angle θ with respect to the original square, rotate the diametral line *in the same direction* through *twice* this angle; i.e. 2θ . The new end points of the line can now be labeled x' and y' , and their σ - τ values are the stresses on the rotated x' - y' axes as shown in Fig. 5(c).

There is nothing mysterious or magical about the Mohr's circle; it is simply a device to help visualize how stresses and other second-rank tensors change when the axes are rotated.

It is clear in looking at the Mohr's circle in Fig. 5(c) that there is something special about axis rotations that cause the diametral line to become either horizontal or vertical. In the first case, the normal stresses assume maximal values and the shear stresses are zero. These normal stresses are known as the *principal* stresses, σ_{p1} and σ_{p2} , and the planes on which they act are the *principal* planes. If the material is prone to fail by tensile cracking, it will do so by cracking along the principal planes when the value of σ_{p1} exceeds the tensile strength.

Example 3

It is instructive to use a Mohr's circle construction to predict how a piece of blackboard chalk will break in torsion, and then verify it in practice. The torsion produces a state of pure shear as shown in Fig. 6,

which causes the principal planes to appear at $\pm 45^\circ$ to the chalk's long axis. The crack will appear transverse to the principal tensile stress, producing a spiral-like failure surface. (As the crack progresses into the chalk, the state of pure shear is replaced by a more complicated stress distribution, so the last part of the failure surface deviates from this ideal path to one running along the axial direction.) This is the same type of fracture that occurred all too often in skiers' femurs, before the advent of modern safety bindings.

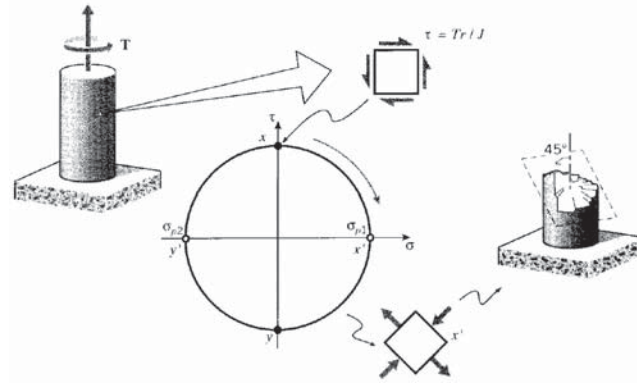


Figure 6: Mohr's circle for simple torsion.

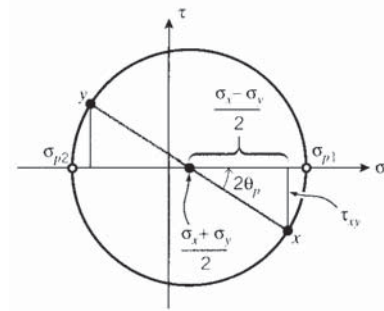


Figure 7: Principal stresses on Mohr's circle.

By direct Pythagorean construction as shown in Fig. 7, the Mohr's circle shows that the angle from the x - y axes to the principal planes is

$$\tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2} \quad (10)$$

and the values of the principal stresses are

$$\sigma_{p1,p1} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad (11)$$

where the first term above is the σ -coordinate of the circle's center, and the second is its radius.

When the Mohr's circle diametral line is vertical, the shear stresses become maximum, equal in magnitude to the radius of the circle:

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \frac{\sigma_{p1} - \sigma_{p2}}{2} \quad (12)$$

The points of maximum shear are 90° away from the principal stress points on the Mohr's circle, so on the actual specimen the planes of maximum shear are 45° from the principal planes. The molecular sliding associated with yield is driven by shear, and usually takes place on the planes of maximum shear. A tensile specimen has principal planes along and transverse to its loading direction, so shear slippage will occur on planes $\pm 45^\circ$ from the loading direction. These slip planes can often be observed as “shear bands” on the specimen.

Note that normal stresses may appear on the planes of maximum shear, so the situation is not quite the converse of the principal planes, on which the shear stresses vanish while the normal stresses are maximum. If the normal stresses happen to vanish on the planes of maximum shear, the stress state is said to be one of “pure shear,” such as is induced by simple torsion. A state of pure shear is therefore one for which a rotation of axes exists such that the normal stresses vanish, which is possible only if the center of the Mohr's circle is at the origin, i.e. $(\sigma_x + \sigma_y)/2 = 0$. More generally, a state of pure shear is one in which the trace of the stress (and strain) matrix vanishes.

Example 4

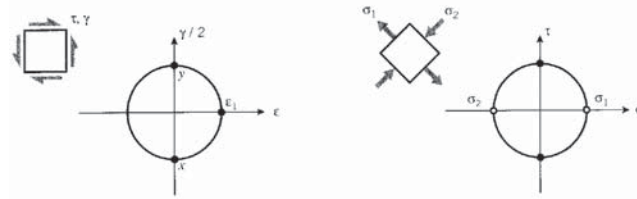


Figure 8: Strain and stress Mohr's circles for simple shear.

Mohr's circles can be drawn for strains as well as stresses, with shear strain plotted on the ordinate and normal strain on the abscissa. However, the ordinate must be $\gamma/2$ rather than just γ , due to the way classical infinitesimal strains are defined. Consider a state of pure shear with strain γ and stress τ as shown in Fig. 8, such as might be produced by placing a circular shaft in torsion. A Mohr's circle for strain quickly shows the principal strain, on a plane 45° away, is given by $\epsilon_1 = \gamma/2$. Hooke's law for shear gives $\tau = G\gamma$, so $\epsilon_1 = \tau/2G$. The principal strain is also related to the principal stresses by

$$\epsilon_1 = \frac{1}{E}(\sigma_1 - \nu\sigma_2)$$

The Mohr's circle for stress gives $\sigma_1 = -\sigma_2 = \tau$, so this can be written

$$\frac{\tau}{2G} = \frac{1}{E}[\tau - \nu(-\tau)]$$

Canceling τ and rearranging, we have the relation among elastic constants stated earlier without proof:

$$G = \frac{E}{2(1 + \nu)}$$

General approach

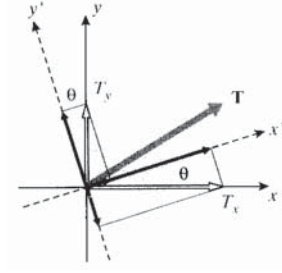


Figure 9: Transformation of vectors.

Another approach to the stress transformation equations, capable of easy extension to three dimensions, starts with the familiar relations by which *vectors* are transformed in two dimensions (see Fig. 9):

$$T_{x'} = T_x \cos \theta + T_y \sin \theta$$

$$T_{y'} = -T_x \sin \theta + T_y \cos \theta$$

In matrix form, this is

$$\begin{Bmatrix} T_{x'} \\ T_{y'} \end{Bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{Bmatrix} T_x \\ T_y \end{Bmatrix}$$

or

$$\mathbf{T}' = \mathbf{a}\mathbf{T} \quad (13)$$

where $\mathbf{a}$ is another transformation matrix that serves to transform the vector components in the original coordinate system to those in the primed system. In index-notation terms, this could also be denoted a_{ij} , so that

$$T'_i = a_{ij}T_j$$

The individual elements of a_{ij} are the cosines of the angles between the i^{th} primed axis and the j^{th} unprimed axis.

It can be shown by direct examination that the $\mathbf{a}$ matrix has the useful property that its inverse equals its transpose; i.e., $\mathbf{a}^{-1} = \mathbf{a}^T$. We can multiply Eqn. 13 by $\mathbf{a}^T$ to give

$$\mathbf{a}^T\mathbf{T}' = (\mathbf{a}^T\mathbf{a})\mathbf{T} = \mathbf{T} \quad (14)$$

so the transformation can go from primed to unprimed, or the reverse.

These relations can be extended to yield an expression for transformation of stresses (or strains, or moments of inertia, or other similar quantities). Recall Cauchy's relation in matrix form:

$$[\boldsymbol{\sigma}]\hat{\mathbf{n}} = \mathbf{T}$$

Using Eqn. 14 to transform the $\hat{\mathbf{n}}$ and $\mathbf{T}$ vectors into their primed counterparts, we have

$$[\boldsymbol{\sigma}] \mathbf{a}^T \hat{\mathbf{n}}' = \mathbf{a}^T \mathbf{T}'$$

Multiplying through by $\mathbf{a}$:

$$(\mathbf{a}[\boldsymbol{\sigma}]\mathbf{a}^T) \hat{\mathbf{n}}' = (\mathbf{a}\mathbf{a}^T) \mathbf{T}' = \mathbf{T}'$$

This is just Cauchy's relation again, but in the primed coordinate frame. The quantity in parentheses must therefore be $[\boldsymbol{\sigma}']$:

$$[\boldsymbol{\sigma}'] = \mathbf{a}[\boldsymbol{\sigma}]\mathbf{a}^T \quad (15)$$

Therefore, transformation of stresses can be done by pre- and postmultiplying by the same transformation matrix applicable to vector transformation. This can also be written out using index notation, which provides another illustration of the transformation differences between scalars (zero-rank tensors), vectors (first-rank tensors), and second-rank tensors:

$$\begin{aligned} \text{rank 0:} & \quad b' = b \\ \text{rank 1:} & \quad T'_i = a_{ij}T_j \\ \text{rank 2:} & \quad \sigma'_{ij} = a_{ij}a_{kl}\sigma_{kl} \end{aligned} \quad (16)$$

In practical work, it is not always a simple matter to write down the nine elements of the $\mathbf{a}$ matrix needed in Eqn. 15. The squares of the components of $\hat{\mathbf{n}}$ for any given plane must sum to unity, and in order for the three planes of the transformed stress cube to be mutually perpendicular the dot product between any two plane normals must vanish. So not just any nine numbers will make sense. Obtaining $\mathbf{a}$ is made much easier by using “Euler angles” to describe axis transformations in three dimensions.

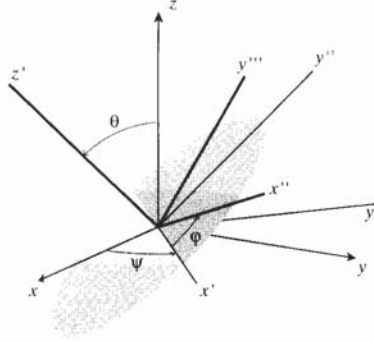


Figure 10: Transformation in terms of Euler angles.

As shown in Fig. 10, the final transformed axes are visualized as being achieved in three steps: first, rotate the original x - y - z axes by an angle ψ (psi) around the z -axis to obtain a new frame we may call x' - y' - z . Next, rotate this new frame by an angle θ about the x' axis to obtain another frame we can call x' - y'' - z' . Finally, rotate this frame by an angle ϕ (phi) around the z' axis to obtain the final frame x'' - y''' - z' . These three transformations correspond to the transformation matrix

$$\mathbf{a} = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This multiplication would certainly be a pain if done manually, but is a natural for a computational approach.

Example 5

The output below shows a computer evaluation of a three-dimensional stress transformation, in this case using MapleTM symbolic mathematics software.

```
# read linear algebra library
> with(linalg):
# Define Euler-angle transformation matrices:
> a1:=array(1..3,1..3,[[cos(psi),sin(psi),0],[-sin(psi),cos(psi),0],[0,0
> ,1]]);
               [cos(psi)   sin(psi)   0]
          a1 := [-sin(psi)  cos(psi)   0]
               [    0         0       1]
> a2:=array(1..3,1..3,[[1,0,0],[0,cos(theta),sin(theta)],[0,-sin(theta),
> cos(theta)]]);
               [1      0      0]
          a2 := [0   cos(theta)  sin(theta)]
               [0  -sin(theta)  cos(theta)]
> a3:=array(1..3,1..3,[[cos(phi),sin(phi),0],[-sin(phi),cos(phi),0],[0,0
> ,1]]);
               [cos(phi)   sin(phi)   0]
          a3 := [-sin(phi)  cos(phi)   0]
               [    0         0       1]
# Overall transformation matrix (multiply individual Euler matrices):
> a:=a1&a2&a3;
          a := (a1 & a2) & a3
# Set precision and read in Euler angles (converted to radians); here
# we are rotating 30 degrees around the z axis only.
> Digits:=4;psi:=0;theta:=30*(Pi/180);phi:=0;
          Digits := 4
          psi := 0
          theta := 1/6 Pi
          phi := 0
# Display transformation matrix for these angles: "evalf" evaluates the
# matrix element, and "map" applies the evaluation to each element of
# the matrix.
> aa:=map(evalf,evalm(a));
          [1.      0.      0. ]
          aa := [0.      .8660   .5000]
               [0.     -.5000   .8660]
# Define the stress matrix in the unprimed frame:
> sigma:=array(1..3,1..3,[[1,2,3],[2,4,5],[3,5,6]]);
               [1  2  3]
          sigma := [2  4  5]
               [3  5  6]
# The stress matrix in the primed frame is then given by Eqn. 15:
> 'sigma_prime'=map(evalf,evalm(aa&sigma&transpose(aa)));
          [ 1.      3.232   1.598]
sigma_prime = [3.232   8.830   3.366]
               [1.598   3.366   1.170]
```

Principal stresses and planes in three dimensions

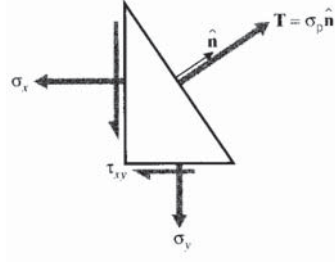


Figure 11: Traction vector normal to principal plane.

The Mohr's circle procedure is not capable of finding principal stresses for three-dimensional stress states, and a more general method is needed. In three dimensions, we seek orientations of axes such that no shear stresses appear, leaving only normal stresses in three orthogonal directions. The vanishing of shear stresses on a plane means that the stress vector $\mathbf{T}$ is normal to the plane, illustrated in two dimensions in Fig. 11. The traction vector can therefore be written as

$$\mathbf{T} = \sigma_p \hat{\mathbf{n}}$$

where σ_p is a simple scalar quantity, the *magnitude* of the stress vector. Using this in Cauchy's relation:

$$\boldsymbol{\sigma} \hat{\mathbf{n}} = \mathbf{T} = \sigma_p \hat{\mathbf{n}}$$

$$(\boldsymbol{\sigma} - \sigma_p \mathbf{I}) \hat{\mathbf{n}} = \mathbf{0} \quad (17)$$

Here $\mathbf{I}$ is the unit matrix. This system will have a nontrivial solution ($\hat{\mathbf{n}} \neq \mathbf{0}$) only if its determinant is zero:

$$|\boldsymbol{\sigma} - \sigma_p \mathbf{I}| = \begin{vmatrix} \sigma_x - \sigma_p & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y - \sigma_p & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z - \sigma_p \end{vmatrix} = 0$$

Expanding the determinant yields a cubic polynomial equation in σ_p :

$$f(\sigma_p) = \sigma_p^3 - I_1 \sigma_p^2 + I_2 \sigma_p - I_3 = 0 \quad (18)$$

This is the *characteristic equation* for stress, where the coefficients are

$$I_1 = \sigma_x + \sigma_y + \sigma_z = \sigma_{kk} \quad (19)$$

$$I_2 = \sigma_x \sigma_y + \sigma_x \sigma_z + \sigma_y \sigma_z - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{xz}^2 = \frac{1}{2} (\sigma_{ii} \sigma_{jj} - \sigma_{ij} \sigma_{ij}) \quad (20)$$

$$I_3 = \det |\sigma| = \frac{1}{3} \sigma_{ij} \sigma_{jk} \sigma_{ki} \quad (21)$$

These I parameters are known as the *invariants* of the stress state; they do not change with transformation of the coordinates and can be used to characterize the overall nature of the stress. For instance I_1 , which has been identified earlier as the trace of the stress matrix, will be seen in a later section to be a measure of the tendency of the stress state to induce hydrostatic dilation or compression. We have already noted that the stress state is one of pure shear if its trace vanishes.

Since the characteristic equation is cubic in σ_p , it will have three roots, and it can be shown that all three roots must be real. These roots are just the principal stresses σ_{p1} , σ_{p2} , and σ_{p3} .

Example 6

Consider a state of simple shear with $\tau_{xy} = 1$ and all other stresses zero:

$$[\sigma] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The invariants are

$$I_1 = 0, \quad I_2 = -1, \quad I_3 = 0$$

and the characteristic equation is

$$\sigma_p^3 - \sigma_p = 0$$

This equation has roots of (-1,0,1) corresponding to principal stresses $\sigma_{p1} = 1$, $\sigma_{p2} = 0$, $\sigma_{p3} = -1$, and is plotted in Fig. 12. This is the same stress state considered in Example 4, and the roots of the characteristic equation agree with the principal values shown by the Mohr's circle.

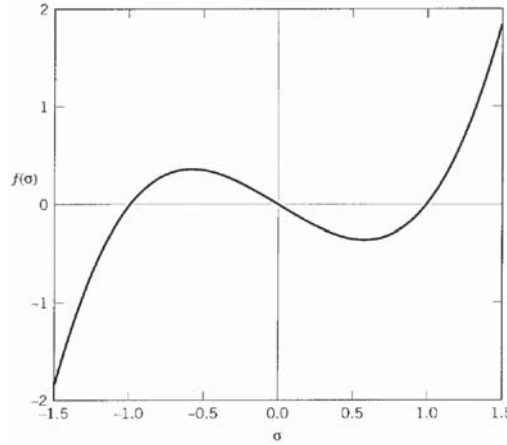


Figure 12: The characteristic equation for $\tau_{xy} = 1$, all other stresses zero.

Problems

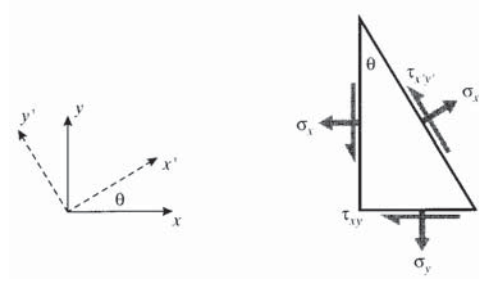
1. Develop an expression for the stress needed to cause transverse failure in a unidirectionally oriented composite as a function of the angle between the load direction and the fiber direction, and show this function in a plot of strength versus θ .
2. Use a free-body force balance to derive the two-dimensional Cartesian stress transformation equations as

$$\begin{aligned}\sigma_{x'} &= \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta \\ \sigma_{y'} &= \sigma_x \sin^2 \theta + \sigma_y \cos^2 \theta - 2\tau_{xy} \sin \theta \cos \theta \\ \tau_{x'y'} &= (\sigma_y - \sigma_x) \sin \theta \cos \theta + \tau_{xy}(\cos^2 \theta - \sin^2 \theta)\end{aligned}$$

Or

$$\begin{Bmatrix} \sigma_{x'} \\ \sigma_{y'} \\ \tau_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix}$$

where $c = \cos \theta$ and $s = \sin \theta$.



Prob. 2

3. Develop mathematical relations for displacements and gradients along transformed axes of the form

$$u' = u \cos \theta + v \sin \theta$$

$$\frac{\partial}{\partial x'} = \frac{\partial}{\partial x} \cdot \frac{\partial x}{\partial x'} + \frac{\partial}{\partial y} \cdot \frac{\partial y}{\partial x'} = \frac{\partial}{\partial x} \cdot \cos \theta + \frac{\partial}{\partial y} \cdot \sin \theta$$

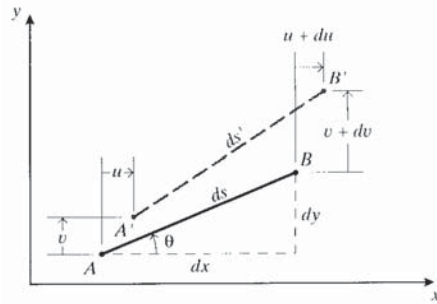
with analogous expressions for v' and $\partial/\partial y'$. Use these to obtain the strain transformation equations (Eqn. 6).

4. Consider a line segment AB of length $ds^2 = dx^2 + dy^2$, oriented at an angle θ from the Cartesian $x - y$ axes as shown. Let the differential displacement of end B relative to end A be

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

Use this geometry to derive the strain transformation equations (Eqn. 6), where the x' axis is along line AB .



Prob. 4

5. Employ double-angle trigonometric relations to show that the two-dimensional Cartesian stress transformation equations can be written in the form

$$\begin{aligned}\sigma_{x'} &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \\ \tau_{x'y'} &= -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \\ \sigma_{y'} &= \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta\end{aligned}$$

Use these relations to justify the Mohr's circle construction.

6. Use matrix multiplication (Eqns. 5 or 8) to transform the following stress and strain states to axes rotated by $\theta = 30^\circ$ from the original x - y axes.

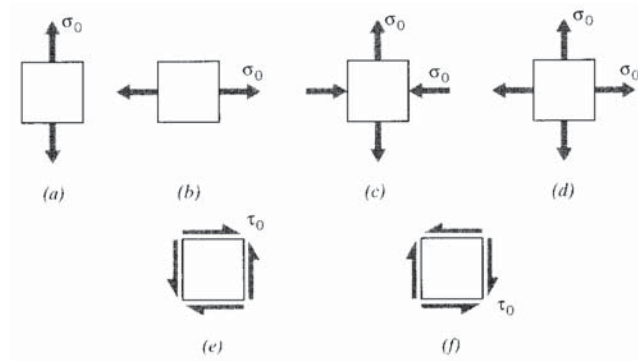
(a)

$$\boldsymbol{\sigma} = \begin{Bmatrix} 1.0 \\ -2.0 \\ 3.0 \end{Bmatrix}$$

(b)

$$\boldsymbol{\epsilon} = \begin{Bmatrix} 0.01 \\ -0.02 \\ 0.03 \end{Bmatrix}$$

7. Sketch the Mohr's circles for each of the stress states shown in the figure below.
8. Construct Mohr's circle solutions for the transformations of Prob. 6.
9. Draw the Mohr's circles and determine the magnitudes of the principal stresses for the following stress states. Denote the principal stress state on a suitably rotated stress square.
- (a) $\sigma_x = 30$ MPa, $\sigma_y = -10$ MPa, $\tau_{xy} = 25$ MPa.
- (b) $\sigma_x = -30$ MPa, $\sigma_y = -90$ MPa, $\tau_{xy} = -40$ MPa.
- (c) $\sigma_x = -10$ MPa, $\sigma_y = 20$ MPa, $\tau_{xy} = -15$ MPa.



Prob. 7

10. Show that the values of principal stresses given by Mohr's circle agree with those obtained mathematically by setting to zero the derivatives of the stress with respect to the transformation angle.
11. For the 3-dimensional stress state $\sigma_x = 25$, $\sigma_y = -15$, $\sigma_z = -30$, $\tau_{yz} = 20$, $\tau_{xz} = 10$, $\tau_{xy} = 30$ (all in MPa):
 - (a) Determine the stress state for Euler angles $\psi = 20^\circ$, $\theta = 30^\circ$, $\phi = 25^\circ$.
 - (b) Plot the characteristic equation.
 - (c) Determine the principal stresses.

Constitutive Equations

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Introduction

The modules on kinematics (Module 8), equilibrium (Module 9), and tensor transformations (Module 10) contain concepts vital to Mechanics of Materials, but they do not provide insight on the role of the material itself. The kinematic equations relate strains to displacement gradients, and the equilibrium equations relate stress to the applied tractions on loaded boundaries and also govern the relations among stress gradients within the material. In three dimensions there are six kinematic equations and three equilibrium equations, for a total of nine. However, there are fifteen variables: three displacements, six strains, and six stresses. We need six more equations, and these are provided by the material's *constitutive* relations: six expressions relating the stresses to the strains. These are a sort of mechanical equation of state, and describe how the material is *constituted mechanically*.

With these constitutive relations, the vital role of the material is reasserted: The elastic constants that appear in this module are material properties, subject to control by processing and microstructural modification as outlined in Module 2. This is an important tool for the engineer, and points up the necessity of considering design *of* the material as well as *with* the material.

Isotropic elastic materials

In the general case of a linear relation between components of the strain and stress tensors, we might propose a statement of the form

$$\epsilon_{ij} = S_{ijkl} \sigma_{kl}$$

where the S_{ijkl} is a *fourth-rank* tensor. This constitutes a sequence of nine equations, since each component of ϵ_{ij} is a linear combination of all the components of σ_{ij} . For instance:

$$\epsilon_{23} = S_{2311} \sigma_{11} + S_{2312} \sigma_{12} + \cdots + S_{2333} \epsilon_{33}$$

Based on each of the indices of S_{ijkl} taking on values from 1 to 3, we might expect a total of 81 independent components in S . However, both ϵ_{ij} and σ_{ij} are symmetric, with six rather than nine independent components each. This reduces the number of S components to 36, as can be seen from a linear relation between the pseudovector forms of the strain and stress:

$$\begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{16} \\ S_{21} & S_{22} & \cdots & S_{26} \\ \vdots & \vdots & \ddots & \vdots \\ S_{61} & S_{26} & \cdots & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{Bmatrix} \quad (1)$$

It can be shown¹ that the $\mathbf{S}$ matrix in this form is also symmetric. It therefore it contains only 21 independent elements, as can be seen by counting the elements in the upper right triangle of the matrix, including the diagonal elements ($1 + 2 + 3 + 4 + 5 + 6 = 21$).

If the material exhibits symmetry in its elastic response, the number of independent elements in the $\mathbf{S}$ matrix can be reduced still further. In the simplest case of an *isotropic* material, whose stiffnesses are the same in all directions, only *two* elements are independent. We have earlier shown that in two dimensions the relations between strains and stresses in isotropic materials can be written as

$$\boxed{\begin{aligned} \epsilon_x &= \frac{1}{E} (\sigma_x - \nu \sigma_y) \\ \epsilon_y &= \frac{1}{E} (\sigma_y - \nu \sigma_x) \\ \gamma_{xy} &= \frac{1}{G} \tau_{xy} \end{aligned}} \quad (2)$$

along with the relation

$$G = \frac{E}{2(1 + \nu)}$$

Extending this to three dimensions, the pseudovector-matrix form of Eqn. 1 for isotropic materials is

$$\begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E} & \frac{-\nu}{E} & \frac{-\nu}{E} & 0 & 0 & 0 \\ \frac{-\nu}{E} & \frac{1}{E} & \frac{-\nu}{E} & 0 & 0 & 0 \\ \frac{-\nu}{E} & \frac{-\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G} \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{Bmatrix} \quad (3)$$

The quantity in brackets is called the *compliance matrix* of the material, denoted $\mathbf{S}$ or S_{ij} . It is important to grasp the physical significance of its various terms. Directly from the rules of matrix multiplication, the element in the i^{th} row and j^{th} column of S_{ij} is the contribution of the j^{th} stress to the i^{th} strain. For instance the component in the 1,2 position is the contribution of the y -direction stress to the x -direction strain: multiplying σ_y by $1/E$ gives the y -direction strain generated by σ_y , and then multiplying this by $-\nu$ gives the Poisson strain induced in the x direction. The zero elements show the lack of coupling between the normal and shearing components.

The isotropic constitutive law can also be written using index notation as (see Prob. 1)

$$\epsilon_{ij} = \frac{1 + \nu}{E} \sigma_{ij} - \frac{\nu}{E} \delta_{ij} \sigma_{kk} \quad (4)$$

where here the indicial form of strain is used and G has been eliminated using $G = E/2(1 + \nu)$. The symbol δ_{ij} is the *Kronecker delta*, described in the Module on Matrix and Index Notation.

¹G.M. Mase, *Schaum's Outline of Theory and Problems of Continuum Mechanics*, McGraw-Hill, 1970.

If we wish to write the stresses in terms of the strains, Eqns. 3 can be inverted. In cases of plane stress ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$), this yields

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix} \quad (5)$$

where again G has been replaced by $E/2(1+\nu)$. Or, in abbreviated notation:

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\epsilon} \quad (6)$$

where $\mathbf{D} = \mathbf{S}^{-1}$ is the *stiffness matrix*.

Hydrostatic and distortional components

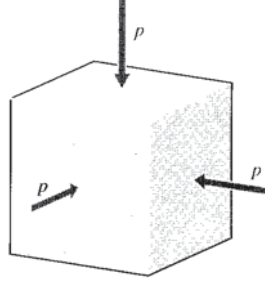


Figure 1: Hydrostatic compression.

A state of hydrostatic compression, depicted in Fig. 1, is one in which no shear stresses exist and where all the normal stresses are equal to the hydrostatic pressure:

$$\sigma_x = \sigma_y = \sigma_z = -p$$

where the minus sign indicates that compression is conventionally positive for pressure but negative for stress. For this stress state it is obviously true that

$$\frac{1}{3}(\sigma_x + \sigma_y + \sigma_z) = \frac{1}{3}\sigma_{kk} = -p$$

so that the hydrostatic pressure is the negative *mean normal stress*. This quantity is just one third of the stress invariant I_1 , which is a reflection of hydrostatic pressure being the same in all directions, not varying with axis rotations.

In many cases other than direct hydrostatic compression, it is still convenient to “dissociate” the hydrostatic (or “dilatational”) component from the stress tensor:

$$\sigma_{ij} = \frac{1}{3}\sigma_{kk}\delta_{ij} + \Sigma_{ij} \quad (7)$$

Here Σ_{ij} is what is left over from σ_{ij} after the hydrostatic component is subtracted. The Σ_{ij} tensor can be shown to represent a state of pure shear, i.e. there exists an axis transformation such that all normal stresses vanish (see Prob. 5). The Σ_{ij} is called the distortional, or deviatoric,

component of the stress. Hence all stress states can be thought of as having two components as shown in Fig. 2, one purely extensional and one purely distortional. This concept is convenient because the material response to these stress components is very different ways. For instance, plastic and viscous flow is driven dominantly by distortional components, with the hydrostatic component causing only elastic deformation.

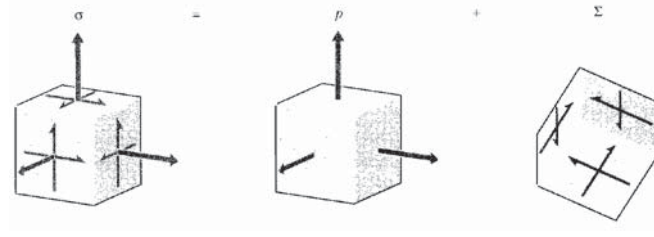


Figure 2: Dilatational and deviatoric components of the stress tensor.

Example 1

Consider the stress state

$$\sigma = \begin{bmatrix} 5 & 6 & 7 \\ 6 & 8 & 9 \\ 7 & 9 & 2 \end{bmatrix}, \text{ GPa}$$

The mean normal stress is $\sigma_{kk}/3 = (5 + 8 + 2)/3 = 5$, so the stress decomposition is

$$\sigma = \frac{1}{3} \sigma_{kk} \delta_{ij} + \Sigma_{ij} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} + \begin{bmatrix} 0 & 6 & 7 \\ 6 & 3 & 9 \\ 7 & 9 & -3 \end{bmatrix}$$

It is not obvious that the deviatoric component given in the second matrix represents pure shear, since there are nonzero components on its diagonal. However, a stress transformation using Euler angles $\psi = \phi = 0, \theta = -9.22^\circ$ gives the stress state

$$\Sigma' = \begin{bmatrix} 0.00 & 4.80 & 7.87 \\ 4.80 & 0.00 & 9.49 \\ 7.87 & 9.49 & 0.00 \end{bmatrix}$$

The hydrostatic component of stress is related to the volumetric strain through the modulus of compressibility ($-p = K \Delta V/V$), so

$$\frac{1}{3} \sigma_{kk} = K \epsilon_{kk} \quad (8)$$

Similarly to the stress, the strain can also be dissociated as

$$\epsilon_{ij} = \frac{1}{3} \epsilon_{kk} \delta_{ij} + e_{ij}$$

where e_{ij} is the deviatoric component of strain. The deviatoric components of stress and strain are related by the material's shear modulus:

$$\Sigma_{ij} = 2G e_{ij} \quad (9)$$

where the factor 2 is needed because tensor descriptions of strain are half the classical strains for which values of G have been tabulated. Writing the constitutive equations in the form of Eqns. 8 and 9 produces a simple form without the coupling terms in the conventional E - ν form.

Example 2

Using the stress state of the previous example along with the elastic constants for steel ($E = 207$ GPa, $\nu = 0.3$, $K = E/3(1 - 2\nu) = 173$ GPa, $G = E/2(1 + \nu) = 79.6$ GPa), the dilatational and distortional components of strain are

$$\delta_{ij}\epsilon_{kk} = \frac{\delta_{ij}\sigma_{kk}}{3K} = \begin{bmatrix} 0.0289 & 0 & 0 \\ 0 & 0.0289 & 0 \\ 0 & 0 & 0.0289 \end{bmatrix}$$

$$e_{ij} = \frac{\Sigma_{ij}}{2G} = \begin{bmatrix} 0 & 0.0378 & 0.0441 \\ 0.0378 & 0.0189 & 0.0567 \\ 0.0441 & 0.0567 & -0.0189 \end{bmatrix}$$

The total strain is then

$$\epsilon_{ij} = \frac{1}{3}\epsilon_{kk}\delta_{ij} + e_{ij} = \begin{bmatrix} 0.00960 & 0.0378 & 0.0441 \\ 0.0378 & 0.0285 & 0.0567 \\ 0.0441 & 0.0567 & -0.00930 \end{bmatrix}$$

If we evaluate the total strain using Eqn. 4, we have

$$\epsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\delta_{ij}\sigma_{kk} = \begin{bmatrix} 0.00965 & 0.0377 & 0.0440 \\ 0.0377 & 0.0285 & 0.0565 \\ 0.0440 & 0.0565 & -0.00915 \end{bmatrix}$$

These results are the same, differing only by roundoff error.

Finite strain model

When deformations become large, geometrical as well as material nonlinearities can arise that are important in many practical problems. In these cases the analyst must employ not only a different strain measure, such as the Lagrangian strain described in Module 8, but also different stress measures (the “Second Piola-Kirchoff stress” replaces the Cauchy stress when Lagrangian strain is used) and different stress-strain constitutive laws as well. A treatment of these formulations is beyond the scope of these modules, but a simple nonlinear stress-strain model for rubbery materials will be outlined here to illustrate some aspects of finite strain analysis. The text by Bathe² provides a more extensive discussion of this area, including finite element implementations.

In the case of small displacements, the strain ϵ_x is given by the expression:

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

For the case of elastomers with $\nu = 0.5$, this can be rewritten in terms of the mean stress $\sigma_m = (\sigma_x + \sigma_y + \sigma_z)/3$ as:

$$2\epsilon_x = \frac{3}{E}(\sigma_x - \sigma_m)$$

²K.-J. Bathe, *Finite Element Procedures in Engineering Analysis*, Prentice-Hall, 1982.

For the large-strain case, the following analogous stress-strain relation has been proposed:

$$\lambda_x^2 = 1 + 2\epsilon_x = \frac{3}{E}(\sigma_x - \sigma_m^*) \quad (10)$$

where here ϵ_x is the Lagrangian strain and σ_m^* is a parameter not necessarily equal to σ_m . The σ_m^* parameter can be found for the case of uniaxial tension by considering the transverse contractions $\lambda_y = \lambda_z$:

$$\lambda_y^2 = \frac{3}{E}(\sigma_y - \sigma_m^*)$$

Since for rubber $\lambda_x \lambda_y \lambda_z = 1$, $\lambda_y^2 = 1/\lambda_x$. Making this substitution and solving for σ_m^* :

$$\sigma_m^* = \frac{-E\lambda_y^2}{3} = \frac{-E}{3\lambda_x}$$

Substituting this back into Eqn. 10,

$$\lambda_x^2 = \frac{3}{E} \left[\sigma_x - \frac{E}{3\lambda_x} \right]$$

Solving for σ_x ,

$$\sigma_x = \frac{E}{3} \left(\lambda_x^2 - \frac{1}{\lambda_x} \right)$$

Here the stress $\sigma_x = F/A$ is the “true” stress based on the actual (contracted) cross-sectional area. The “engineering” stress $\sigma_e = F/A_0$ based on the original area $A_0 = A\lambda_x$ is:

$$\sigma_e = \frac{\sigma_x}{\lambda_x} = G \left(\lambda_x - \frac{1}{\lambda_x^2} \right)$$

where $G = E/2(1 + \nu) = E/3$ for $\nu = 1/2$. This result is the same as that obtained in Module 2 by considering the force arising from the reduced entropy as molecular segments spanning crosslink sites are extended. It appears here from a simple hypothesis of stress-strain response, using a suitable measure of finite strain.

Anisotropic materials

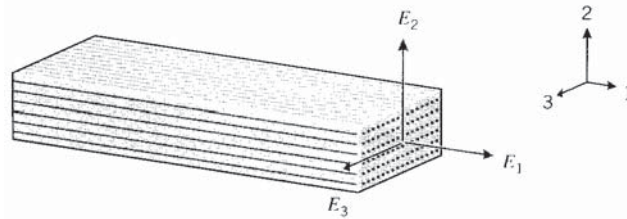


Figure 3: An orthotropic material.

If the material has a texture like wood or unidirectionally-reinforced fiber composites as shown in Fig. 3, the modulus E_1 in the fiber direction will typically be larger than those in the transverse directions (E_2 and E_3). When $E_1 \neq E_2 \neq E_3$, the material is said to be *orthotropic*.

It is common, however, for the properties in the plane transverse to the fiber direction to be isotropic to a good approximation ($E_2 = E_3$); such a material is called *transversely isotropic*. The elastic constitutive laws must be modified to account for this anisotropy, and the following form is an extension of Eqn. 3 for transversely isotropic materials:

$$\begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} 1/E_1 & -\nu_{21}/E_2 & 0 \\ -\nu_{12}/E_1 & 1/E_2 & 0 \\ 0 & 0 & 1/G_{12} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} \quad (11)$$

The parameter ν_{12} is the *principal Poisson's ratio*; it is the ratio of the strain induced in the 2-direction by a strain applied in the 1-direction. This parameter is not limited to values less than 0.5 as in isotropic materials. Conversely, ν_{21} gives the strain induced in the 1-direction by a strain applied in the 2-direction. Since the 2-direction (transverse to the fibers) usually has much less stiffness than the 1-direction, it should be clear that a given strain in the 1-direction will usually develop a much larger strain in the 2-direction than will the same strain in the 2-direction induce a strain in the 1-direction. Hence we will usually have $\nu_{12} > \nu_{21}$. There are five constants in the above equation (E_1 , E_2 , ν_{12} , ν_{21} and G_{12}). However, only four of them are independent; since the S matrix is symmetric, $\nu_{21}/E_2 = \nu_{12}/E_1$.

A table of elastic constants and other properties for widely used anisotropic materials can be found in the Module on Composite Ply Properties.

The simple form of Eqn. 11, with zeroes in the terms representing coupling between normal and shearing components, is obtained only when the axes are aligned along the *principal material directions*; i.e. along and transverse to the fiber axes. If the axes are oriented along some other direction, all terms of the compliance matrix will be populated, and the symmetry of the material will not be evident. If for instance the fiber direction is off-axis from the loading direction, the material will develop shear strain as the fibers try to orient along the loading direction as shown in Fig. 4. There will therefore be a coupling between a normal stress and a shearing strain, which never occurs in an isotropic material.

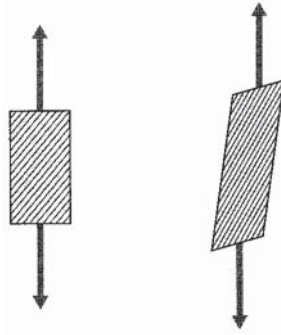


Figure 4: Shear-normal coupling.

The transformation law for compliance can be developed from the transformation laws for strains and stresses, using the procedures described in Module 10 (Transformations). By successive transformations, the pseudovector form for strain in an arbitrary x - y direction shown in Fig. 5 is related to strain in the 1-2 (principal material) directions, then to the stresses in the 1-2 directions, and finally to the stresses in the x - y directions. The final grouping of transformation matrices relating the x - y strains to the x - y stresses is then the transformed compliance matrix

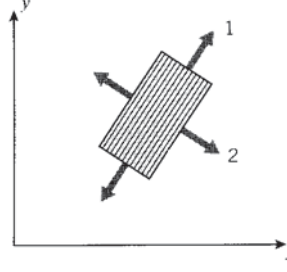


Figure 5: Axis transformation for constitutive equations.

in the x - y direction:

$$\begin{aligned} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix} &= \mathbf{R} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{1}{2}\gamma_{12} \end{Bmatrix} = \mathbf{R}\mathbf{A}^{-1} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{1}{2}\gamma_{12} \end{Bmatrix} = \mathbf{R}\mathbf{A}^{-1}\mathbf{R}^{-1} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{Bmatrix} \\ &= \mathbf{R}\mathbf{A}^{-1}\mathbf{R}^{-1}\mathbf{S} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \mathbf{R}\mathbf{A}^{-1}\mathbf{R}^{-1}\mathbf{S}\mathbf{A} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} \equiv \bar{\mathbf{S}} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} \end{aligned}$$

where $\bar{\mathbf{S}}$ is the *transformed compliance matrix* relative to x - y axes. Here $\mathbf{A}$ is the transformation matrix, and $\mathbf{R}$ is the Reuter's matrix defined in the Module on Tensor Transformations. The inverse of $\bar{\mathbf{S}}$ is $\bar{\mathbf{D}}$, the stiffness matrix relative to x - y axes:

$$\boxed{\bar{\mathbf{S}} = \mathbf{R}\mathbf{A}^{-1}\mathbf{R}^{-1}\mathbf{S}\mathbf{A}, \quad \bar{\mathbf{D}} = \bar{\mathbf{S}}^{-1}} \quad (12)$$

Example 3

Consider a ply of Kevlar-epoxy composite with a stiffnesses $E_1 = 82$, $E_2 = 4$, $G_{12} = 2.8$ (all GPa) and $\nu_{12} = 0.25$. The compliance matrix $\mathbf{S}$ in the 1-2 (material) direction is:

$$\mathbf{S} = \begin{bmatrix} 1/E_1 & -\nu_{21}/E_2 & 0 \\ -\nu_{12}/E_1 & 1/E_2 & 0 \\ 0 & 0 & 1/G_{12} \end{bmatrix} = \begin{bmatrix} .1220 \times 10^{-10} & -.3050 \times 10^{-11} & 0 \\ -.3050 \times 10^{-11} & .2500 \times 10^{-9} & 0 \\ 0 & 0 & .3571 \times 10^{-9} \end{bmatrix}$$

If the ply is oriented with the fiber direction (the “1” direction) at $\theta = 30^\circ$ from the x - y axes, the appropriate transformation matrix is

$$\mathbf{A} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} = \begin{bmatrix} .7500 & .2500 & .8660 \\ .2500 & .7500 & -.8660 \\ -.4330 & .4330 & .5000 \end{bmatrix}$$

The compliance matrix relative to the x - y axes is then

$$\bar{\mathbf{S}} = \mathbf{R}\mathbf{A}^{-1}\mathbf{R}^{-1}\mathbf{S}\mathbf{A} = \begin{bmatrix} .8830 \times 10^{-10} & -.1970 \times 10^{-10} & -.1222 \times 10^{-9} \\ -.1971 \times 10^{-10} & .2072 \times 10^{-9} & -.8371 \times 10^{-10} \\ -.1222 \times 10^{-9} & -.8369 \times 10^{-10} & -.2905 \times 10^{-9} \end{bmatrix}$$

Note that this matrix is symmetric (to within numerical roundoff error), but that nonzero coupling values exist. A user not aware of the internal composition of the material would consider it completely anisotropic.

The apparent engineering constants that would be observed if the ply were tested in the x - y rather than 1-2 directions can be found directly from the transformed $\bar{\mathbf{S}}$ matrix. For instance, the apparent elastic modulus in the x direction is $E_x = 1/\bar{\mathbf{S}}_{1,1} = 1/(.8830 \times 10^{-10}) = 11.33 \text{ GPa}$.

Problems

1. Expand the indicial forms of the governing equations for solid elasticity in three dimensions:

$$\text{equilibrium : } \sigma_{ij,j} = 0$$

$$\text{kinematic : } \epsilon_{ij} = (u_{i,j} + u_{j,i})/2$$

$$\text{constitutive : } \epsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\delta_{ij}\sigma_{kk} + \alpha\delta_{ij}\Delta T$$

where α is the coefficient of linear thermal expansion and ΔT is a temperature change.

2. (a) Write out the compliance matrix $\mathbf{S}$ of Eqn. 3 for polycarbonate using data in the Module on Material Properties.
 (b) Use matrix inversion to obtain the stiffness matrix $\mathbf{D}$.
 (c) Use matrix multiplication to obtain the stresses needed to induce the strains

$$\boldsymbol{\epsilon} = \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} 0.02 \\ 0.0 \\ 0.03 \\ 0.01 \\ 0.025 \\ 0.0 \end{Bmatrix}$$

3. (a) Write out the compliance matrix $\mathbf{S}$ of Eqn.3 for an aluminum alloy using data in the Module on Material Properties.
 (b) Use matrix inversion to obtain the stiffness matrix $\mathbf{D}$.
 (c) Use matrix multiplication to obtain the stresses needed to induce the strains

$$\boldsymbol{\epsilon} = \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} 0.01 \\ 0.02 \\ 0.0 \\ 0.0 \\ 0.15 \\ 0.0 \end{Bmatrix}$$

4. Given the stress tensor

$$\sigma_{ij} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{bmatrix} \quad (\text{MPa})$$

- (a) Dissociate σ_{ij} into deviatoric and dilatational parts Σ_{ij} and $(1/3)\sigma_{kk}\delta_{ij}$.

(b) Given $G = 357$ MPa and $K = 1.67$ GPa, obtain the deviatoric and dilatational strain tensors e_{ij} and $(1/3)\epsilon_{kk}\delta_{ij}$.

(c) Add the deviatoric and dilatational strain components obtained above to obtain the total strain tensor ϵ_{ij} .

(d) Compute the strain tensor ϵ_{ij} using the alternate form of the elastic constitutive law for isotropic elastic solids:

$$\epsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\delta_{ij}\sigma_{kk}$$

Compare the result with that obtained in (c).

5. Provide an argument that any stress matrix having a zero trace can be transformed to one having only zeroes on its diagonal; i.e. the deviatoric stress tensor Σ_{ij} represents a state of pure shear.
6. Write out the x - y two-dimensional compliance matrix $\bar{\mathbf{S}}$ and stiffness matrix $\bar{\mathbf{D}}$ (Eqn. 12) for a single ply of graphite/epoxy composite with its fibers aligned along the x axes.
7. Write out the x - y two-dimensional compliance matrix $\bar{\mathbf{S}}$ and stiffness matrix $\bar{\mathbf{D}}$ (Eqn. 12) for a single ply of graphite/epoxy composite with its fibers aligned 30° from the x axis.